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JEE MAIN · CHEMISTRY · CHAPTER-WISE PYQS

Chemical Bonding & Molecular Structure

JEE Main Chemistry · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer)	238 questions
Numerical	64 questions
MCQ (More than One Correct Answer)	1 question
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MCQ (Single Correct Answer) 238 questions

Q1. JEE Main 2026 (Online) 8th April Evening Shift

Bromine trifluoride autoionizes to form $BrF_2^{\oplus}$ and $BrF_4^{\ominus}$. The shapes of the cation and anion are respectively ____ and ____ .

Open solution

(A) bent, square planar
(B) linear, square planar
(C) bent, see-saw
(D) linear, tetrahedral

Q2. JEE Main 2026 (Online) 6th April Evening Shift

Given below are two statements :

Statement I : The number of compounds among SO_2, SO_3, SF_4, SF_6 and H_2S in which sulphur does not obey the Octet rule is 3 .

Statement II : Among $[H_2O, ClF_3, SF_4]$, $[NH_3, BrF_5, SF_4]$, $[BrF_5, ClF_3, XeF_4]$ and $[XeF_4, ClF_3, H_2O]$, the number of sets in which all the molecules have one lone pair of electrons on the central atom is 1 .

In the light of the above statements, choose the correct answer from the options given below :

Open solution

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

Q3. JEE Main 2026 (Online) 6th April Evening Shift

Given below are two statements :

Statement I : $F_2O < H_2O < Cl_2O$ is the correct trend in terms of bond angle.

Statement II : SiF_4, SnF_4 and PbF_4 are ionic in nature.

In the light of the above statements, choose the correct answer from the options given below :

Open solution

(A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

Q4. JEE Main 2026 (Online) 6th April Morning Shift

The pairs among

$A = [SO_3^{2-}, CO_3^{2-}], B = [O_2^{2-}, F_2], C = [CN^-, CO], D = [NH_3, H_3O^+]$ and $E = [MnO_4^{2-}, CrO_4^{2-}]$ that do not have similar Lewis dot structure are

Open solution

(A) A, B and E
(B) A and E
(C) B, C and D
(D) C and D

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Q5. JEE Main 2026 (Online) 5th April Evening Shift

Open solution

The covalent radii of atoms A and B are r_A and r_B, respectively. The covalent bond length and total length of AB molecule are respectively

(A) $(\mathrm{r_A}+\mathrm{r_B}), 2\,(\mathrm{r_A}+\mathrm{r_B})$

(B) $\frac{1}{2}(\mathrm{r_A}+\mathrm{r_B}), (\mathrm{r_A}+\mathrm{r_B})$

(C) $(\mathrm{r_A}+\mathrm{r_B}), (\mathrm{r_A}+\mathrm{r_B})$

(D) $2\,(\mathrm{r_A}+\mathrm{r_B}), \frac{1}{2}(\mathrm{r_A}+\mathrm{r_B})$

Q6. JEE Main 2026 (Online) 4th April Evening Shift

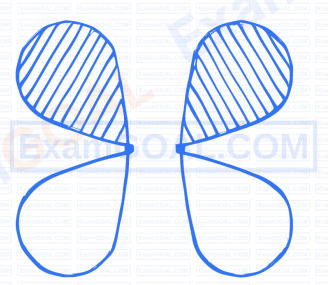
Open solution

Which of the following pictorial diagrams most correctly represents the π^* (π antibonding) molecular orbital between two atoms if the internuclear axis is taken to be in the z-direction ($\xrightarrow{z\text{-axis}}$) ?

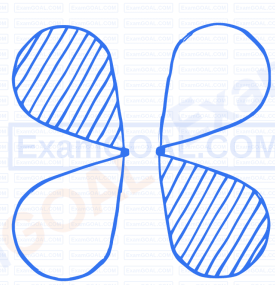
(A)



(B)



(C)



(D)



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Q7. JEE Main 2026 (Online) 2nd April Evening Shift

Open solution

SF_4 is isostructural with :

A. $BrF_4^{\ominus}$

B. CH_4

C. $IF_4^{\oplus}$

D. XeF_4

E. XeO_2F_2

Choose the **correct** answer from the options given below :

(A) C Only **(B)** C and E Only **(C)** A and D Only **(D)** B and E Only

Q8. JEE Main 2026 (Online) 2nd April Morning Shift

Open solution

Given below are two statements :

Statement (I): The correct sequence of bond lengths in the following species is :

$O_2^+ < O_2 < O_2^- < O_2^{2-}$

Statement (II): The correct sequence of number of unpaired electrons in the following species is :

$O_2 > O_2^+ > O_2^- > O_2^{2-}$

In the light of the above statements, choose the **correct answer** from the options given below :

(A) Both **Statement I** and **Statement II** are true

(B) Both **Statement I** and **Statement II** are false

(C) **Statement I** is true but **Statement II** is false

(D) **Statement I** is false but **Statement II** is true

Q9. JEE Main 2026 (Online) 28th January Evening Shift

Open solution

Match List - I with List - II according to shape.

List – I	List – II
A. XeO_3	I. BrF_5
B. XeF_2	II. NH_3
C. XeO_2F_2	III. $[I_3]^-$
D. $XeOF_4$	IV. SF_4

Choose the **correct** answer from the options given below :

(A) A-II, B-III, C-I, D-IV **(B)** A-III, B-II, C-IV, D-I

(C) A-II, B-III, C-IV, D-I **(D)** A-II, B-I, C-III, D-IV

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Q10. JEE Main 2026 (Online) 28th January Morning Shift

Open solution

Given below are two statements :

Statement I : The number of species among BF_4^-, SiF_4, XeF_4 and SF_4, that have unequal $E-F$ bond lengths is two. Here, E is the central atom.

Statement II : Among O_2^-, O_2^{2-}, F_2 and O_2^+, O_2^- has the highest bond order.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are true

Q11. JEE Main 2026 (Online) 24th January Evening Shift

Open solution

Pair of species among the following having same bond order as well as paramagnetic character will be-

(A) O_2^-, N_2^+

(B) $O_2{}^+, N_2{}^{2-}$

(C) O_2^-, N_2^-

(D) O_2^+, N_2^-

Q12. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Given below are statements about some molecules/ions.

Identify the CORRECT statements.

A. The dipole moment value of NF_3 is higher than that of NH_3.

B. The dipole moment value of BeH_2 is zero.

C. The bond order of $O_2{}^{2-}$ and F_2 is same.

D. The formal charge on the central oxygen atom of ozone is -1 .

E. In NO_2, all the three atoms satisfy the octet rule, hence it is very stable.

Choose the correct answer from the options given below:

(A) B & C Only (B) B, C & D Only (C) A, C & D Only (D) A, B, C, D & E

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Q13. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Among the following, the CORRECT combinations are

A. $IF_3 \rightarrow$ T-shaped ($sp^3 d$)

B. $IF_5 \rightarrow$ Square pyramidal $\left(sp^3 d^2\right)$

C. $IF_7 \rightarrow$ Pentagonal bipyramidal $\left(sp^3 d^3\right)$

D. $ClO_4^- \rightarrow$ Square planar ($sp^2 d$)

Choose the correct answer from the options given below:

(A) B, C and D Only (B) A, B, C and D (C) A, B and C Only (D) A and B Only

Q14. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

Which statements are **NOT TRUE** about XeO_2F_2?

A. It has a see-saw shape.

B. Xe has 5 electron pairs in its valence shell in XeO_2F_2.

C. The $O-Xe-O$ bond angle is close to 180°.

D. The $F-Xe-F$ bond angle is close to 180°.

E. Xe has 16 valence electrons in XeO_2F_2.

Choose the ***correct*** answer from the options given below :

(A) B, D and E Only (B) B and D Only (C) B, C and E Only (D) A and D Only

Q15. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Identify the molecule (X) with maximum number of lone pairs of electrons (obtained using Lewis dot structure) among HNO_3, H_2SO_4, NF_3 and O_3. Choose the correct bond angle made by the central atom of the molecule (X).

(A) 107° (B) 102° (C) 120° (D) 116°

Q16. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

Among H_2S, H_2O, NF_3, NH_3 and $CHCl_3$, identify the molecule (X) with lowest dipole moment value. The number of lone pairs of electrons present on the central atom of the molecule (X) is :

(A) 3 (B) 2 (C) 0 (D) 1

Q17. JEE Main 2026 (Online) 22nd January Morning Shift

Open solution

Two p-block elements X and Y form fluorides of the type EF_3. The fluoride compound XF_3 is a Lewis acid and YF_3 is a Lewis base. The hybridizations of the central atoms of XF_3 and YF_3 respectively are

(A) sp^2 and sp^3 (B) Both sp^3

(C) Both sp^2 (D) sp^3 and sp^2

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Q18. JEE Main 2026 (Online) 22nd January Morning Shift

The formal charges on the atoms marked as (1) to (4) in the Lewis representation of HNO_3 molecule respectively are



Open solution

(A) $0, 0, -1, +1$ (B) $0, -1, 0, +1$ (C) $0, +1, 0, -1$ (D) $+1, 0, 0, -1$

Q19. JEE Main 2026 (Online) 21st January Evening Shift

The correct increasing order of C–H(A), C–O(B), C=O(C) and C≡N(D) bonds in terms of covalent bond length is :

(A) D < C < B < A (B) D < C < A < B (C) A < B < C < D (D) A < D < C < B

Open solution

Q20. JEE Main 2026 (Online) 21st January Evening Shift

Given below are two statements :

Statement I : The correct order in terms of bond dissociation enthalpy is $Cl_2 > Br_2 > F_2 > I_2$.

Statement II : The correct trend in the covalent character of the metal halides is [$SnCl_4 > SnCl_2$], [$PbCl_4 > PbCl_2$] and [$UF_4 > UF_6$].

In the light of the above statements, choose the **correct** answer from the options given below :

(A) Both **Statement I** and **Statement II** are true

(B) Both **Statement I** and **Statement II** are false

(C) **Statement I** is false but **Statement II** is true

(D) **Statement I** is true but **Statement II** is false

Open solution

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Q21. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

Given below are two statements :

Statement I :

The number of species among $SF_4, NH_4^+, [NiCl_4]^{2-}, XeF_4, [PtCl_4]^{2-}, SeF_4$ and $[Ni(CN)_4]^{2-}$, that have tetrahedral geometry is 3 .

Statement II :

In the set [$NO_2, BeH_2, BF_3, AlCl_3$], all the molecules have incomplete octet around central atom.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q22. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

In SO_2, NO_2^- and N_3^-, the hybridizations at the central atom are respectively :

(A) sp^2, sp^2 and sp^2 (B) sp, sp^2 and sp (C) sp^2, sp and sp (D) sp^2, sp^2 and sp

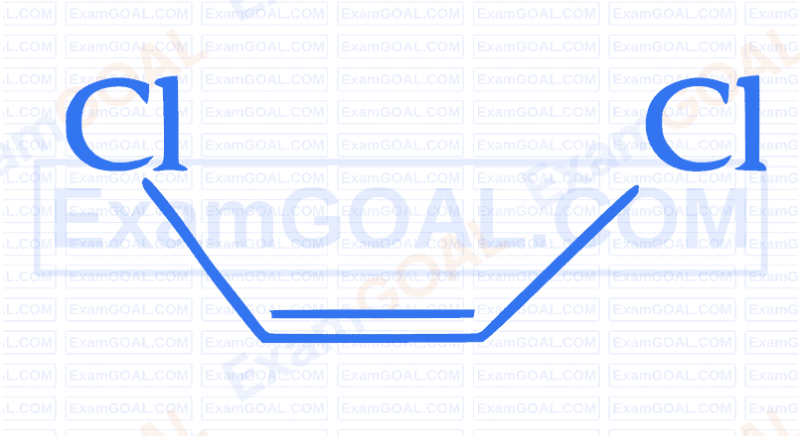
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Q23. JEE Main 2025 (Online) 7th April Evening Shift

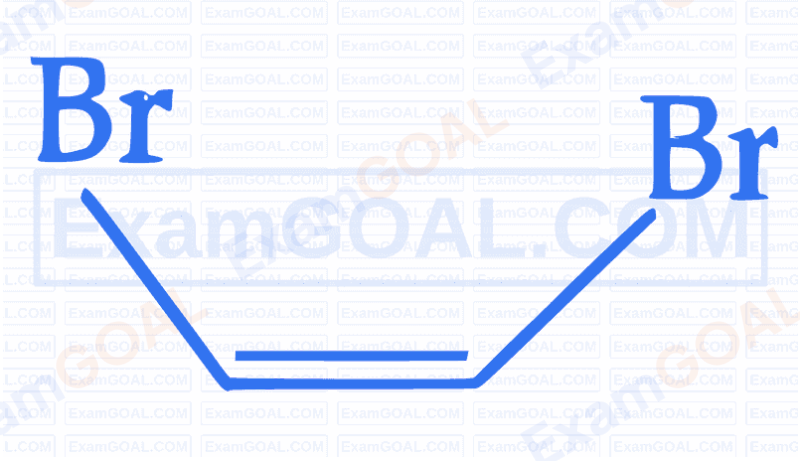
Open solution

Given below are two statements :

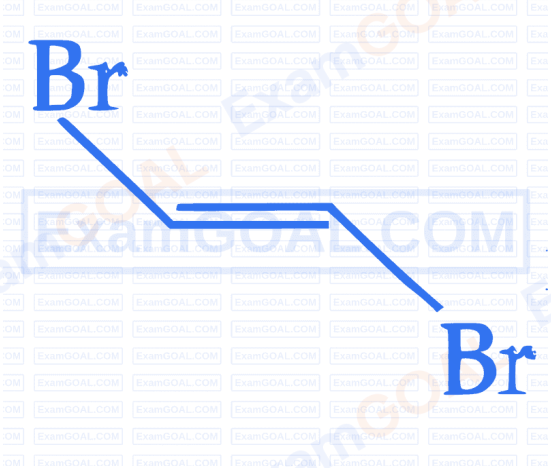
Statement (I) :



is more polar than

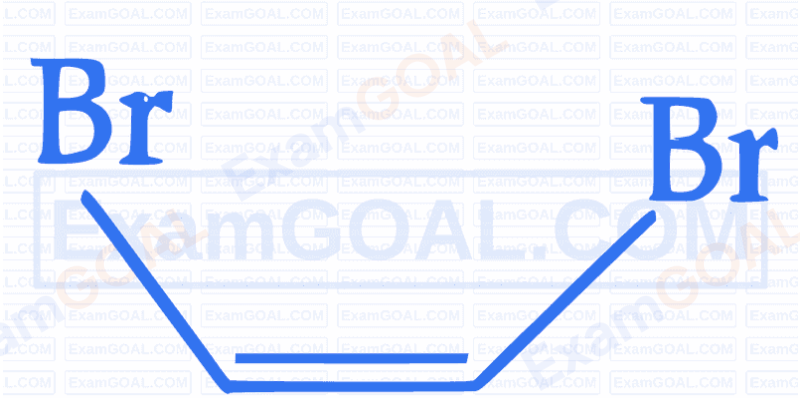


Statement (II) : Boiling point of



is lower than

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but it is more polar than



.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(A) **Both Statement I and Statement II are correct**

(B) **Statement I is incorrect but Statement II is correct**

(C) **Both Statement I and Statement II are incorrect**

(D) **Statement I is correct but Statement II is incorrect**

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Q24. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

Match the List I with List II.

	List - I Molecule/Ion		List - II Bond pair : lone pair (on the central atom)
(A)	ICl_2^-	(I)	$4:2$
(B)	H_2O	(II)	$4:1$
(C)	SO_2	(III)	$2:3$
(D)	XeF_4	(IV)	$2:2$

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-II, D-I

(B) A-III, B-IV, C-I, D-II

(C) A-II, B-I, C-IV, D-III

(D) A-III, B-IV, C-II, D-I

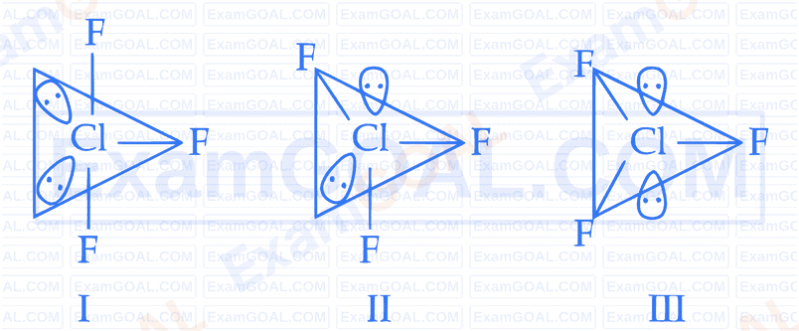
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Q25. JEE Main 2025 (Online) 4th April Evening Shift

Open solution

Given below are two statements:

Statement (I): For  , all three possible structures may be drawn as follows.



Statement (II): Structure III is most stable, as the orbitals having the lone pairs are axial, where the lp – bp repulsion is minimum.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

(A) Both Statement I and Statement II are correct

(B) Both Statement I and Statement II are incorrect

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

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Q26. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Given below are two statements.

Statement I: The dipole moment of $\overset{4}{CH_3}-\overset{3}{CH}=\overset{2}{CH}-\overset{1}{CH}=O$ is greater than $\overset{4}{CH_3}-\overset{3}{CH_2}-\overset{2}{CH_2}-\overset{1}{CH}=O$

Statement II: C_1-C_2 bond length of $\overset{4}{CH_3}-\overset{3}{CH}=\overset{2}{CH}-\overset{1}{CH}=O$ is greater than C_1-C_2 bond length of $\overset{4}{CH_3}-\overset{3}{CH_2}-\overset{2}{CH_2}-\overset{1}{CH}=O$

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q27. JEE Main 2025 (Online) 4th April Morning Shift

Open solution

Which of the following molecule(s) show/s paramagnetic behavior?

A. O_2

B. N_2

C. F_2

D. S_2

E. Cl_2

Choose the correct answer from the options given below:

(A) A & C only (B) A & E Only (C) A & D Only (D) B Only

Q28. JEE Main 2025 (Online) 3rd April Evening Shift

Open solution

Given below are two statements:

Statement I : Wet cotton clothes made of cellulose based carbohydrate takes comparatively longer time to get dried than wet nylon polymer based clothes.

Statement II : Intermolecular hydrogen bonding with water molecule is more in nylon-based clothes than in the case of cotton clothes.

In the light of above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are false

(B) Both Statement I and Statement II are true

(C) Statement I is false but Statement II is true

(D) Statement I is true but Statement II is false

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Q29. JEE Main 2025 (Online) 2nd April Evening Shift

Which among the following molecules is (a) involved in sp^3d hybridization, (b) has different bond lengths and (c) has lone pair of electrons on the central atom?

(A) XeF_4 (B) XeF_2 (C) PF_5 (D) SF_4

Open solution

Q30. JEE Main 2025 (Online) 2nd April Morning Shift

A molecule with the formula AX_4Y has all its elements from p-block. Element A is rarest, monoatomic, non-radioactive from its group and has the lowest ionization enthalpy value among A, X and Y . Elements X and Y have first and second highest electronegativity values respectively among all the known elements. The shape of the molecule is:

(A) Pentagonal planar (B) Square pyramidal

(C) Trigonal bipyramidal (D) Octahedral

Open solution

Q31. JEE Main 2025 (Online) 2nd April Morning Shift

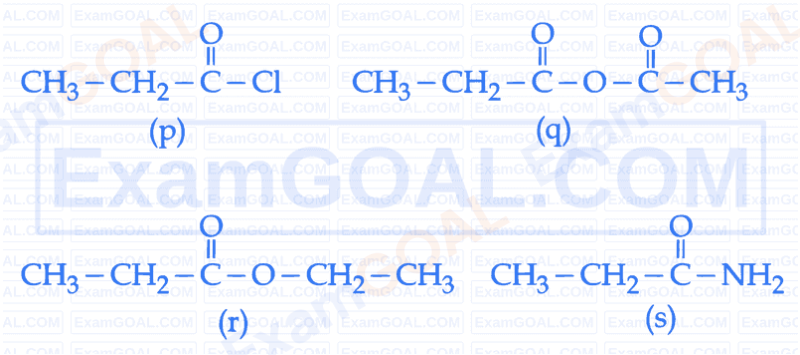
Among $SO_2, NF_3, NH_3, XeF_2, ClF_3$ and SF_4, the hybridization of the molecule with nonzero dipole moment and highest number of lone-pairs of electrons on the central atom is

(A) sp^3 (B) dsp^2 (C) sp^3d^2 (D) sp^3d

Open solution

Q32. JEE Main 2025 (Online) 2nd April Morning Shift

Consider the following molecules:



The correct order of rate of hydrolysis is :

(A) $p > q > r > s$ (B) $r > q > p > s$

(C) $p > r > q > s$ (D) $q > p > r > s$

Open solution

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Q33. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Consider ' n ' is the number of lone pair of electrons present in the equatorial position of the most stable structure of ClF_3. The ions from the following with ' n ' number of unpaired electrons are

A. V^{3+}

B. Ti^{3+}

C. Cu^{2+}

D. Ni^{2+}

E. Ti^{2+}

Choose the correct answer from the options given below:

(A) B and D Only (B) A, D and E Only (C) B and C Only (D) A and C Only

Q34. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

The molecules having square pyramidal geometry are

(A) BrF_5 & PCl_5 (B) SbF_5 & PCl_5

(C) BrF_5 & $XeOF_4$ (D) SbF_5 & $XeOF_4$

Q35. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Given below are two statements :

Statement (I) : Experimentally determined oxygen-oxygen bond lengths in the O_3 are found to be same and the bond length is greater than that of a $O=O$ (double bond) but less than that of a single $(O-O)$ bond.

Statement (II) : The strong lone pair-lone pair repulsion between oxygen atoms is solely responsible for the fact that the bond length in ozone is smaller than that of a double bond $(O=O)$ but more than that of a single bond $(O-O)$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Statement I is false but Statement II is true

(B) Both Statement I and Statement II are true

(C) Statement I is true but Statement II is false

(D) Both Statement I and Statement II are false

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Q36. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

Which of the following linear combination of atomic orbitals will lead to formation of molecular orbitals in homonuclear diatomic molecules [internuclear axis in z-direction] ?

A. $2p_z$ and $2p_x$

B. 2 s and $2p_x$

C. $3d_{xy}$ and $3d_{x^2-y^2}$

D. 2 s and $2p_z$

E. $2p_z$ and $3d_{x^2-y^2}$

Choose the correct answer from the options given below:

(A) D Only **(B)** E Only **(C)** C and D Only **(D)** A and B Only

Q37. JEE Main 2025 (Online) 24th January Morning Shift

Open solution

Which of the following statement is true with respect to H_2O, NH_3 and CH_4 ?

A. The central atoms of all the molecules are sp^3 hybridized.

B. The $H-O-H, H-N-H$ and $H-C-H$ angles in the above molecules are $104.5^\circ, 107.5^\circ$ and 109.5°, respectively.

C. The increasing order of dipole moment is $CH_4 < NH_3 < H_2O$.

D. Both H_2O and NH_3 are Lewis acids and CH_4 is a Lewis base.

E. A solution of NH_3 in H_2O is basic. In this solution NH_3 and H_2O act as Lowry-Bronsted acid and base respectively.

Choose the correct answer from the options given below:

(A) A, D and E Only **(B)** C, D and E Only

(C) A, B and C Only **(D)** A, B, C and E Only

Q38. JEE Main 2025 (Online) 23rd January Morning Shift

Open solution

Match the List - I with List - II

	List - I (Classification of molecules based on octet rule)		**List - II (Example)**
(A)	Molecules obeying octet rule	(I)	NO, NO_2
(B)	Molecules with incomplete octet	(II)	$BCl_3, AlCl_3$
(C)	Molecules with incomplete octet with odd electron	(III)	H_2SO_4, PCl_5
(D)	Molecules with expanded octet	(IV)	CCl_4, CO_2

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II **(B)** A-II, B-IV, C-III, D-I

(C) A-IV, B-II, C-I, D-III **(D)** A-III, B-II, C-I, D-IV

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Q39. JEE Main 2025 (Online) 22nd January Evening Shift

Open solution

Arrange the following compounds in increasing order of their dipole moment :

HBr, H_2S, NF_3 and $CHCl_3$

(A) $HBr < H_2S < NF_3 < CHCl_3$

(B) $H_2S < HBr < NF_3 < CHCl_3$

(C) $NF_3 < HBr < H_2S < CHCl_3$

(D) $CHCl_3 < NF_3 < HBr < H_2S$

Q40. JEE Main 2024 (Online) 9th April Evening Shift

Open solution

The correct increasing order for bond angles among BF_3, PF_3 and ClF_3 is :

(A) $BF_3 = PF_3 < ClF_3$

(B) $BF_3 < PF_3 < ClF_3$

(C) $ClF_3 < PF_3 < BF_3$

(D) $PF_3 < BF_3 < ClF_3$

Q41. JEE Main 2024 (Online) 9th April Morning Shift

Open solution

In which one of the following pairs the central atoms exhibit sp^2 hybridization ?

(A) BF_3 and NO_2^-

(B) NH_2^- and BF_3

(C) NH_2^- and H_2O

(D) H_2O and NO_2

Q42. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

The shape of carbocation is :

(A) tetrahedral

(B) diagonal pyramidal

(C) diagonal

(D) trigonal planar

Q43. JEE Main 2024 (Online) 8th April Evening Shift

Open solution

When ψ_A and ψ_B are the wave functions of atomic orbitals, then σ^* is represented by :

(A) $\psi_A + 2\psi_B$

(B) $\psi_A - \psi_B$

(C) $\psi_A - 2\psi_B$

(D) $\psi_A + \psi_B$

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Q44. JEE Main 2024 (Online) 8th April Morning Shift

Open solution

Match List I with List II

	LIST I (Molecule)		LIST II (Shape)
A.	NH_3	I.	Square pyramid
B.	BrF_5	II.	Tetrahedral
C.	PCl_5	III.	Trigonal pyramidal
D.	CH_4	IV.	Trigonal bipyramidal

Choose the correct answer from the options given below:

(A) A-II, B-IV, C-I, D-III

(B) A-III, B-I, C-IV, D-II

(C) A-IV, B-III, C-I, D-II

(D) A-III, B-IV, C-I, D-II

Q45. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

Match List I with List II

	LIST I (Compound/Species)		LIST II (Shape/Geometry)
A.	SF_4	I.	Tetrahedral
B.	BrF_3	II.	Pyramidal
C.	BrO_3^-	III.	See saw
D.	NH_4^+	IV.	Bent T-Shape

Choose the correct answer from the options given below:

(A) A-III, B-II, C-IV, D-I

(B) A-III, B-IV, C-II, D-I

(C) A-II, B-IV, C-III, D-I

(D) A-II, B-III, C-I, D-IV

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Q46. JEE Main 2024 (Online) 6th April Morning Shift

Open solution

Match List I with List II

	LIST I (Molecule/Species)		LIST II (Property/Shape)
A.	SO_2Cl_2	I.	Paramagnetic
B.	NO	II.	Diamagnetic
C.	NO_2^-	III.	Tetrahedral
D.	I_3^-	IV.	Linear

Choose the correct answer from the options given below:

(A) A-II, B-III, C-I, D-IV

(B) A-III, B-I, C-II, D-IV

(C) A-IV, B-I, C-III, D-II

(D) A-III, B-IV, C-II, D-I

Q47. JEE Main 2024 (Online) 5th April Evening Shift

Open solution

Match List I with List II.

	LIST I		LIST II
A.	ICl	I.	T - shape
B.	ICl_3	II.	Square pyramidal
C.	ClF_5	III.	Pentagonal bipyramidal
D.	IF_7	IV.	Linear

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(IV), (C)-(III), (D)-(II)

(B) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(C) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)

(D) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Q48. JEE Main 2024 (Online) 5th April Evening Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : NH_3 and NF_3 molecule have pyramidal shape with a lone pair of electrons on nitrogen atom. The resultant dipole moment of NH_3 is greater than that of NF_3.

Reason (R) : In NH_3, the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the $N-H$ bonds. F is the most electronegative element.

In the light of the above statements, choose the correct answer from the options given below :

(A) (A) is false but (R) is true

(B) (A) is true but (R) is false

(C) Both (A) and (R) are true but (R) is NOT the correct explanation of (A)

(D) Both (A) and (R) are true and (R) is the correct explanation of (A)

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Q49. JEE Main 2024 (Online) 5th April Morning Shift

Number of σ and π bonds present in ethylene molecule is respectively :

(A) 3 and 1 (B) 5 and 1 (C) 4 and 1 (D) 5 and 2

Open solution

Q50. JEE Main 2024 (Online) 4th April Evening Shift

The correct statement/s about Hydrogen bonding is/are

A. Hydrogen bonding exists when H is covalently bonded to the highly electronegative atom.

B. Intermolecular H bonding is present in o-nitro phenol

C. Intramolecular H bonding is present in HF.

D. The magnitude of H bonding depends on the physical state of the compound.

E. H-bonding has powerful effect on the structure and properties of compounds

Choose the correct answer from the options given below:

(A) A, B, C only (B) A only (C) A, D, E only (D) A, B, D only

Open solution

Q51. JEE Main 2024 (Online) 4th April Evening Shift

The number of species from the following that have pyramidal geometry around the central atom is _____

$\mathrm{S_2O_3^{2-}}, \mathrm{SO_4^{2-}}, \mathrm{SO_3^{2-}}, \mathrm{S_2O_7^{2-}}$

(A) 4 (B) 3 (C) 2 (D) 1

Open solution

Q52. JEE Main 2024 (Online) 4th April Morning Shift

Number of molecules/ions from the following in which the central atom is involved in $\mathrm{sp^3}$ hybridization is _____.

$\mathrm{NO_3^-}, \mathrm{BCl_3}, \mathrm{ClO_2^-}, \mathrm{ClO_3^-}$

(A) 2 (B) 4 (C) 1 (D) 3

Open solution

Q53. JEE Main 2024 (Online) 4th April Morning Shift

Which one of the following molecules has maximum dipole moment?

(A) $\mathrm{CH_4}$ (B) $\mathrm{NF_3}$ (C) $\mathrm{NH_3}$ (D) $\mathrm{PF_5}$

Open solution

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Q54. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Given below are two statements :

Statement (I) : A π bonding MO has lower electron density above and below the inter-nuclear axis.

Statement (II) : The π^* antibonding MO has a node between the nuclei.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

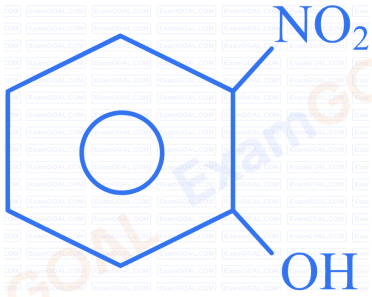
(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q55. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

Select the compound from the following that will show intramolecular hydrogen bonding.

(A) 

(B) H_2O

(C) C_2H_5OH

(D) NH_3

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Q56. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): PH_3 has lower boiling point than NH_3.

Reason (R) : In liquid state NH_3 molecules are associated through van der Waals forces, but PH_3 molecules are associated through hydrogen bonding.

In the light of the above statements, choose the **most appropriate** answer from the options given below :

(A) Both (A) and (R) are correct and (R) is the correct explanation of (A)

(B) (A) is not correct but (R) is correct

(C) (A) is correct but (R) is not correct

(D) Both $(\mathbf{A})$ and $(\mathbf{R})$ are correct but $(\mathbf{R})$ is not the correct explanation of $(\mathbf{A})$

Q57. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

Arrange the bonds in order of increasing ionic character in the molecules. LiF, K_2O, N_2, SO_2 and ClF_3 :

(A) $N_2 < SO_2 < ClF_3 < K_2O < LiF$

(B) $ClF_3 < N_2 < SO_2 < K_2O < LiF$

(C) $LiF < K_2O < ClF_3 < SO_2 < N_2$

(D) $N_2 < ClF_3 < SO_2 < K_2O < LiF$

Q58. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Which of the following is least ionic?

(A) $CoCl_2$ (B) KCl (C) $BaCl_2$ (D) AgCl

Q59. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

The linear combination of atomic orbitals to form molecular orbitals takes place only when the combining atomic orbitals

A. have the same energy

B. have the minimum overlap

C. have same symmetry about the molecular axis

D. have different symmetry about the molecular axis

Choose the most appropriate from the options given below:

(A) B, C, D only (B) A, B, C only (C) B and D only (D) A and C only

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Q60. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Given below are two statements:

Statement - I: Since Fluorine is more electronegative than nitrogen, the net dipole moment of NF_3 is greater than NH_3.

Statement - II: In NH_3, the orbital dipole due to lone pair and the dipole moment of NH bonds are in opposite direction, but in NF_3 the orbital dipole due to lone pair and dipole moments of N-F bonds are in same direction.

In the light of the above statements, choose the most appropriate from the options given below:

(A) Statement I is true but Statement II is false.

(B) Both Statement I and Statement II are true.

(C) Both Statement I and Statement II are false.

(D) Statement I is false but Statement II is true.

Q61. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

The molecule / ion with square pyramidal shape is

(A) PCl_5

(B) $[Ni(CN)_4]^{2-}$

(C) PF_5

(D) BrF_5

Q62. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Aluminium chloride in acidified aqueous solution forms an ion having geometry

(A) Square planar

(B) Octahedral

(C) Trigonal bipyramidal

(D) Tetrahedral

Q63. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Match List I with List II.

	List I Molecule		List II Shape
(A)	BrF_5	(I)	T-Shape
(B)	H_2O	(II)	See saw
(C)	ClF_3	(III)	Bent
(D)	SF_4	(IV)	Square pyramidal

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(II), (C)-(IV), (D)-(III)

(B) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(C) (A)-(III), (B)-(IV), (C)-(I), (D)-(II)

(D) (A)-(II), (B)-(I), (C)-(III), (D)-(IV)

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Q64. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : There is a considerable increase in covalent radius from N to P. However from As to Bi only a small increase in covalent radius is observed.

Reason (R) : Covalent and ionic radii in a particular oxidation state increases down the group. In the light of the above statements, choose the most appropriate answer from the options given below:

(A) (A) is false but (R) is true

(B) Both (A) and (R) are true and (R) is the correct explanation of (A)

(C) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(D) (A) is true but (R) is false

Q65. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

The difference in energy between the actual structure and the lowest energy resonance structure for a given compound is

(A) electromeric energy

(B) resonance energy

(C) ionization energy

(D) hyperconjugation energy

Q66. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

Choose the polar molecule from the following:

(A) $\mathrm{CCl_4}$ **(B)** $\mathrm{CO_2}$ **(C)** $\mathrm{CH_2=CH_2}$ **(D)** $\mathrm{CHCl_3}$

Q67. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

Consider the following statements:

(A) NF_3 molecule has a trigonal planar structure.

(B) Bond length of $\mathrm{N_2}$ is shorter than $\mathrm{O_2}$.

(C) Isoelectronic molecules or ions have identical bond order.

(D) Dipole moment of $\mathrm{H_2S}$ is higher than that of water molecule.

Choose the correct answer from the options given below:

(A) (C) and (D) are correct

(B) (A) and (B) are correct

(C) (B) and (C) are correct

(D) (A) and (D) are correct

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Q68. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

Given below are two statements :

Statement I : SO_2 and H_2O both possess V-shaped structure.

Statement II : The bond angle of SO_2 is less than that of H_2O.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Both Statement I and Statement II are incorrect

(B) Both Statement I and Statement II are correct

(C) Statement I is correct but Statement II is incorrect

(D) Statement I is incorrect but Statement II is correct

Q69. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

In which of the following processes, the bond order increases and paramagnetic character changes to diamagnetic one ?

(A) $O_2 \rightarrow O_2^{2-}$

(B) $N_2 \rightarrow N_2^+$

(C) $NO \rightarrow NO^+$

(D) $O_2 \rightarrow O_2^+$

Q70. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

ClF_5 at room temperature is a:

(A) Colourless liquid with square pyramidal geometry

(B) Colourless gas with square pyramidal geometry

(C) Colourless gas with trigonal bipyramidal geometry.

(D) Colourless liquid with trigonal bipyramidal geometry

Q71. JEE Main 2023 (Online) 12th April Morning Shift

Open solution

The bond order and magnetic property of acetylide ion are same as that of

(A) O_2^+

(B) O_2^-

(C) N_2^+

(D) NO^+

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Q72. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

Match List - I with List - II:

	List - I Species		List - II Geometry/Shape
A.	H_3O^+	I.	Tetrahedral
B.	Acetylide anion	II.	Linear
C.	NH_4^+	III.	Pyramidal
D.	ClO_2^-	IV.	Bent

Choose the correct answer from the options given below:

(A) A-III, B-I, C-II, D-IV **(B)** A-III, B-II, C-I, D-IV

(C) A-III, B-IV, C-I, D-II **(D)** A-III, B-IV, C-II, D-I

Q73. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The compound which does not exist is

(A) $(NH_4)_2BeF_4$ **(B)** $PbEt_4$ **(C)** BeH_2 **(D)** NaO_2

Q74. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The pair from the following pairs having both compounds with net non-zero dipole moment is

(A) cis-butene, trans-butene **(B)** CH_2Cl_2, $CHCl_3$

(C) Benzene, anisidine **(D)** 1,4-Dichlorobenzene, 1,3-Dichlorobenzene

Q75. JEE Main 2023 (Online) 6th April Morning Shift

Open solution

Match List I with List II

	LIST I Oxide		LIST II Type of bond
A.	N_2O_4	I.	1 N = O bond
B.	NO_2	II.	1 N $-$ O $-$ N bond
C.	N_2O_5	III.	1 N $-$ N bond
D.	N_2O	IV.	1 N=N / N $\equiv$ N bond

Choose the correct answer from the options given below:

(A) A-II, B-I, C-III, D-IV **(B)** A-III, B-I, C-II, D-IV

(C) A-III, B-I, C-IV, D-II **(D)** A-II, B-IV, C-III, D-I

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Q76. JEE Main 2023 (Online) 1st February Evening Shift

Open solution

$O-O$ bond length in H_2O_2 is $\underline{X}$ than the $O-O$ bond length in F_2O_2. The $O-H$ bond length in H_2O_2 is $\underline{Y}$ than that of the $O-F$ bond in F_2O_2.

Choose the correct option for $\underline{X}$ and $\underline{Y}$ from those given below :

(A) X - shorter, Y - longer

(B) X - longer, Y - shorter

(C) X - shorter, Y - shorter

(D) X - longer, Y - longer

Q77. JEE Main 2023 (Online) 31st January Morning Shift

Open solution

Match List I with List II

	List I		List II
A.	XeF_4	I.	See-saw
B.	SF_4	II.	Square-planar
C.	NH_4^+	III.	Bent T-shaped
D.	BrF_3	IV.	Tetrahedral

Choose the correct answer from the options given below :

(A) A - II, B - I, C - III, D - IV

(B) A - II, B - I, C - IV, D - III

(C) A - IV, B - I, C - II, D - III

(D) A - IV, B - III, C - II, D - I

Q78. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

For OF_2 molecule consider the following :

A. Number of lone pairs on oxygen is 2 .

B. FOF angle is less than 104.5°.

C. Oxidation state of O is -2.

D. Molecule is bent 'V' shaped.

E. Molecular geometry is linear.

correct options are:

(A) A, C, D only

(B) A, B, D only

(C) C, D, E only

(D) B, E, A only

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Q79. JEE Main 2023 (Online) 30th January Morning Shift

Open solution

Match List I with List II

	List I (molecules/ions)		List II (No. of lone pairs of e^- on central atom)
A.	IF_7	I.	Three
B.	ICl_4^-	II.	One
C.	XeF_6	III.	Two
D.	XeF_2	IV.	Zero

Choose the correct answer from the options given below :

(A) A - II, B - I, C - IV, D - III

(B) A - IV, B - I, C - II, D - III

(C) A - IV, B - III, C - II, D - I

(D) A - II, B - III, C - IV, D - I

Q80. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

According to MO theory the bond orders for O_2^{2-}, CO and NO^+ respectively, are

(A) 1, 3 and 3

(B) 2, 3 and 3

(C) 1, 2 and 3

(D) 1, 3 and 2

Q81. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

The magnetic behaviour of Li_2O, Na_2O_2 and KO_2, respectively, are :

(A) paramagnetic, paramagnetic and diamagnetic

(B) paramagnetic, diamagnetic and paramagnetic

(C) diamagnetic, paramagnetic and diamagnetic

(D) diamagnetic, diamagnetic and paramagnetic

Q82. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

The bond dissociation energy is highest for

(A) Cl_2

(B) I_2

(C) Br_2

(D) F_2

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Q83. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Statement I : Dipole moment is a vector quantity and by convention it is depicted by a small arrow with tail on the negative centre and head pointing towards the positive centre.

Statement II : The crossed arrow of the dipole moment symbolizes the direction of the shift of charges in the molecules.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is correct but Statement II is incorrect

(B) Statement I is incorrect but Statement II is correct

(C) Both Statement I and Statement II are correct

(D) Both Statement I and Statement II are incorrect

Q84. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

What is the number of unpaired electron(s) in the highest occupied molecular orbital of the following species : $N_2; N_2^+; O_2; O_2^+$?

(A) 0, 1, 0, 1 (B) 2, 1, 0, 1 (C) 0, 1, 2, 1 (D) 2, 1, 2, 1

Q85. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

Decreasing order of the hydrogen bonding in following forms of water is correctly represented by

A. Liquid water

B. Ice

C. Impure water

Choose the correct answer from the options given below :

(A) A = B > C (B) B > A > C (C) A > B > C (D) C > B > A

Q86. JEE Main 2023 (Online) 24th January Morning Shift

Open solution

Order of Covalent bond;

A. $KF > KI; LiF > KF$

B. $KF < KI; LiF > KF$

C. $SnCl_4 > SnCl_2; CuCl > NaCl$

D. $LiF > KF; CuCl < NaCl$

E. $KF < KI; CuCl > NaCl$

Choose the correct answer from the options given below :

(A) B, C only (B) A, B only (C) C, E only (D) B, C, E only

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Q87. JEE Main 2022 (Online) 30th July Evening Shift

The correct option for relative covalent nature of compounds is

(A) $KF > KI$ (B) $SnCl_4 > SnCl_2$ (C) $KF > LiF$ (D) $NaCl > HCl$

Open solution

Q88. JEE Main 2022 (Online) 29th July Morning Shift

Which of the following pair of molecules contains an odd electron molecule and an expanded octet molecule?

(A) BCl_3 and SF_6 (B) NO and H_2SO_4

(C) SF_6 and H_2SO_4 (D) BCl_3 and NO

Open solution

Q89. JEE Main 2022 (Online) 29th July Morning Shift

Number of lone pairs of electrons in the central atom of SCl_2, O_3, ClF_3 and SF_6, respectively, are :

(A) 0, 1, 2 and 2 (B) 2, 1, 2 and 0 (C) 1, 2, 2 and 0 (D) 2, 1, 0 and 2

Open solution

Q90. JEE Main 2022 (Online) 27th July Evening Shift

Match List - I with List - II.

	List - I		List - II
(A)	$\psi_{MO} = \psi_A - \psi_B$	(I)	Dipole moment
(B)	$\mu = Q \times r$	(II)	Bonding molecular orbital
(C)	$\frac{N_b - N_a}{2}$	(III)	Anti-bonding molecular orbital
(D)	$\psi_{MO} = \psi_A + \psi_B$	(IV)	Bond order

Choose the correct answer from the options given below :

(A) $(A) - (II), (B) - (I), (C) - (IV), (D) - (III)$

(B) $(A) - (III), (B) - (IV), (C) - (I), (D) - (II)$

(C) $(A) - (III), (B) - (I), (C) - (IV), (D) - (II)$

(D) $(A) - (III), (B) - (IV), (C) - (II), (D) - (I)$

Open solution

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Q91. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

Given below are two statements.

Statement I: O_2, Cu^{2+}, and Fe^{3+} are weakly attracted by magnetic field and are magnetized in the same direction as magnetic field.

Statement II: $NaCl$ and H_2O are weakly magnetized in opposite direction to magnetic field.

In the light of the above statements, choose the most appropriate answer from the options given below.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Q92. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

Arrange the following in increasing order of their covalent character.

A. CaF_2

B. $CaCl_2$

C. $CaBr_2$

D. CaI_2

Choose the correct answer from the options given below.

(A) B < A < C < D **(B)** A < B < C < D **(C)** A < B < D < C **(D)** A < C < B < D

Q93. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

Match List-I with List-II :

	List I (Compound)		List II (Shape)
(A)	BrF_5	(I)	bent
(B)	$[CrF_6]^{3-}$	(II)	square pyramidal
(C)	O_3	(III)	trigonal bipyramidal
(D)	PCl_5	(IV)	octahedral

Choose the correct answer from the options given below :

(A) $(A)-(I),(B)-(II),(C)-(III),(D)-(IV)$

(B) $(A)-(IV),(B)-(III),(C)-(II),(D)-(I)$

(C) $(A)-(II),(B)-(IV),(C)-(I),(D)-(III)$

(D) $(A)-(III),(B)-(IV),(C)-(II),(D)-(I)$

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Q94. JEE Main 2022 (Online) 25th July Evening Shift

Open solution

Match List I with List II:

	List I (molecule)		List II (hybridization ; shape)
(A)	XeO_3	(I)	sp^3d ; linear
(B)	XeF_2	(II)	sp^3 ; pyramidal
(C)	$XeOF_4$	(III)	sp^3d^3 ; distorted octahedral
(D)	XeF_6	(IV)	sp^3d^2 ; square pyramidal

Choose the correct answer from the options given below:

(A) A-II, B-I, C-IV, D-III

(B) A-II, B-IV, C-III, D-I

(C) A-IV, B-II, C-III, D-I

(D) A-IV, B-II, C-I, D-III

Q95. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

Consider the species CH_4, NH_4^+ and BH_4^-. Choose the correct option with respect to these species.

(A) They are isoelectronic and only two have tetrahedral structures.

(B) They are isoelectronic and all have tetrahedral structures.

(C) Only two are isoelectronic and all have tetrahedral structures.

(D) Only two are isoelectronic and only two have tetrahedral structures.

Q96. JEE Main 2022 (Online) 29th June Evening Shift

Open solution

Number of lone pair(s) of electrons on central atom and the shape BrF_3 molecule respectively, are

(A) 0, triangular planar.

(B) 1, pyramidal.

(C) 2, bent T-shape.

(D) 1, bent T-shape.

Q97. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

In the structure of SF_4, the lone pair of electrons on S is in.

(A) equatorial position and there are two lone pair - bond pair repulsions at 90°.

(B) equatorial position and there are three lone pair - bond pair repulsions at 90°.

(C) axial position and there are three lone pair - bond pair repulsion at 90°.

(D) axial position and there are two lone pair - bond pair repulsion at 90°.

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Q98. JEE Main 2022 (Online) 27th June Evening Shift

Identify the incorrect statement for PCl_5 from the following.

(A) In this molecule, orbitals of phosphorus are assumed to undergo sp^3d hybridization.

(B) The geometry of PCl_5 is trigonal bipyramidal.

(C) PCl_5 has two axial bonds stronger than three equatorial bonds.

(D) The three equatorial bonds of PCl_5 lie in a plane.

Open solution

Q99. JEE Main 2022 (Online) 27th June Evening Shift

The correct order of increasing intermolecular hydrogen bond strength is :

(A) $HCN < H_2O < NH_3$ (B) $HCN < CH_4 < NH_3$ (C) $CH_4 < HCN < NH_3$ (D) $CH_4 < NH_3 < HCN$

Open solution

Q100. JEE Main 2022 (Online) 27th June Morning Shift

Based upon VSEPR theory, match the shape (geometry) of the molecules in List-I with the molecules in List-II and select the most appropriate option.

	List - I (Shape)		List - II (Molecules)
(A)	T-shaped	(I)	XeF_4
(B)	Trigonal planar	(II)	SF_4
(C)	Square planar	(III)	ClF_3
(D)	See-saw	(IV)	BF_3

(A) (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

(B) (A) - (III), (B) - (IV), (C) - (I), (D) - (II)

(C) (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

(D) (A) - (IV), (B) - (III), (C) - (I), (D) - (II)

Open solution

Q101. JEE Main 2022 (Online) 26th June Morning Shift

Consider the ions/molecule

$O_2^+, O_2, O_2^-, O_2^{2-}$

For increasing bond order the correct option is :

(A) $O_2^{2-} < O_2^- < O_2 < O_2^+$ (B) $O_2^- < O_2^{2-} < O_2 < O_2^+$

(C) $O_2^- < O_2^{2-} < O_2^+ < O_2$ (D) $O_2^- < O_2^+ < O_2^{2-} < O_2$

Open solution

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Q102. JEE Main 2022 (Online) 25th June Morning Shift

Bonding in which of the following diatomic molecule(s) become(s) stronger, on the basis of MO Theory, by removal of an electron?

(A) NO

(B) N_2

(C) O_2

(D) C_2

(E) B_2

Choose the most appropriate answer from the options given below :

(A) (A), (B), (C) only

(B) (B), (C), (E) only

(C) (A), (C) only

(D) (D) only

Open solution

Q103. JEE Main 2022 (Online) 25th June Morning Shift

Number of electron deficient molecules among the following

PH_3, B_2H_6, CCl_4, NH_3, LiH and BCl_3 is

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

Open solution

Q104. JEE Main 2022 (Online) 24th June Evening Shift

The correct order of bond orders of ${C_2}^{2-}$, ${N_2}^{2-}$ and ${O_2}^{2-}$ is, respectively

(A) ${C_2}^{2-} < {N_2}^{2-} < {O_2}^{2-}$

(B) ${O_2}^{2-} < {N_2}^{2-} < {C_2}^{2-}$

(C) ${C_2}^{2-} < {O_2}^{2-} < {N_2}^{2-}$

(D) ${N_2}^{2-} < {C_2}^{2-} < {O_2}^{2-}$

Open solution

Q105. JEE Main 2021 (Online) 1st September Evening Shift

Number of paramagnetic oxides among the following given oxides is _____.

Li_2O, CaO, Na_2O_2, KO_2, MgO and K_2O

(A) 1 **(B)** 2 **(C)** 3 **(D)** 0

Open solution

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Q106. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

Match List - I with List - II :

	List - I (Species)		List - II (No. of lone pairs of electrons on the central atom)
(a)	XeF_2	(i)	0
(b)	XeO_2F_2	(ii)	1
(c)	XeO_3F_2	(iii)	2
(d)	XeF_4	(iv)	3

Choose the most appropriate answer from the options given below :

(A) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)

(B) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(C) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)

(D) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

Q107. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

Match items of List-I with those of List-II :

	List - I (Property)		**List - II (Example)**
(a)	Diamagnetism	(i)	MnO
(b)	Ferrimagnetism	(ii)	O_2
(c)	Paramagnetism	(iii)	NaCl
(d)	Antiferromagnetism	(iv)	Fe_3O_4

Choose the most appropriate answer from the options given below :

(A) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(B) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

(C) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(D) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

Q108. JEE Main 2021 (Online) 26th August Evening Shift

Open solution

The bond order and magnetic behaviour of O_2^- ion are respectively :

(A) 1.5 and paramagnetic

(B) 1.5 and diamagnetic

(C) 2 and diamagnetic

(D) 1 and paramagnetic

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Q109. JEE Main 2021 (Online) 25th July Evening Shift

In the following the correct bond order sequence is :

(A) $O_2^{2-} > O_2^+ > O_2^- > O_2$

(B) $O_2^+ > O_2^- > O_2^{2-} > O_2$

(C) $O_2^+ > O_2 > O_2^- > O_2^{2-}$

(D) $O_2 > O_2^- > O_2^{2-} > O_2^+$

Open solution

Q110. JEE Main 2021 (Online) 25th July Evening Shift

Identify the species having one π-bond and maximum number of canonical forms from the following :

(A) SO_3 (B) O_2 (C) SO_2 (D) CO_3^{2-}

Open solution

Q111. JEE Main 2021 (Online) 22th July Evening Shift

Match List-I with List-II :

	List-I (Species)		List-II (Hybrid Orbitals)
(a)	SF_4	(i)	sp^3d^2
(b)	IF_5	(ii)	d^2sp^3
(c)	NO_2^+	(iii)	sp^3d
(d)	NH_4^+	(iv)	sp^3
		(v)	sp

Choose the correct answer from the options given below :

(A) (a)-(i), (b)-(ii), (c)-(v) and (d)-(iii)

(B) (a)-(ii), (b)-(i), (c)-(iv) and (d)-(v)

(C) (a)-(iii), (b)-(i), (c)-(v) and (d)-(iv)

(D) (a)-(iv), (b)-(iii), (c)-(ii) and (d)-(v)

Open solution

Q112. JEE Main 2021 (Online) 20th July Evening Shift

The hybridisations of the atomic orbitals of nitrogen in NO_2^-, NO_2^+ and NH_4^+ respectively are.

(A) sp^3, sp^2 and sp (B) sp, sp^2 and sp^3 (C) sp^3, sp and sp^2 (D) sp^2, sp and sp^3

Open solution

Q113. JEE Main 2021 (Online) 17th March Evening Shift

Amongst the following, the linear species is :

(A) O_3 (B) Cl_2O (C) N_3^- (D) NO_2

Open solution

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Q114. JEE Main 2021 (Online) 17th March Morning Shift

A central atom in a molecule has two lone pairs of electrons and forms three single bonds. The shape of this molecule is :

(A) trigonal pyramidal
(B) T-shaped
(C) see-saw
(D) planar triangular

Open solution

Q115. JEE Main 2021 (Online) 16th March Morning Shift

Given below are two statements : one is labeled as Assertion A and the other is labelled as Reason R :

Assertion A : The H−O−H bond angle in water molecule is 104.5°.

Reason R : The lone pair - lone pair repulsion of electrons is higher than the bond pair - bond pair repulsion.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both A and R are true, and R is the correct explanation of A
(B) A is true but R is false
(C) A is false but R is true
(D) Both A and R are true, but R is not the correct explanation of A

Open solution

Q116. JEE Main 2021 (Online) 26th February Evening Shift

Match List - I with List - II.

	List - I (Molecule)		List - II (Bond order)
(a)	Ne_2	(i)	1
(b)	N_2	(ii)	2
(c)	F_2	(iii)	0
(d)	O_2	(iv)	3

Choose the correct answer from the options given below :

(A) (a) → (i), (b) → (ii), (c) → (iii), (d) → (iv)
(B) (a) → (iv), (b) → (iii), (c) → (ii), (d) → (i)
(C) (a) → (iii), (b) → (iv), (c) → (i), (d) → (ii)
(D) (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

Open solution

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Q117. JEE Main 2021 (Online) 26th February Evening Shift

In $\overset{1}{C}H_2 = \overset{2}{C} = \overset{3}{C}H - \overset{4}{C}H_3$ molecule, the hybridization of carbon 1, 2, 3 and 4 respectively, are :

(A) sp^2, sp, sp^2, sp^3

(B) sp^2, sp^2, sp^2, sp^3

(C) sp^3, sp, sp^3, sp^3

(D) sp^2, sp^3, sp^2, sp^3

Open solution

Q118. JEE Main 2021 (Online) 26th February Morning Shift

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation.

Reason R : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are true and R is the correct explanation of A

(B) A is true but R is false

(C) A is false but R is true

(D) Both A and R are true but R is NOT the correct explanation of A

Open solution

Q119. JEE Main 2021 (Online) 25th February Evening Shift

Which among the following species has unequal bond lengths?

(A) XeF_4

(B) BF_4^-

(C) SiF_4

(D) SF_4

Open solution

Q120. JEE Main 2021 (Online) 25th February Morning Shift

According to molecular orbital theory, the species among the following that does not exist is :

(A) O_2^{2-}

(B) Be_2

(C) He_2^-

(D) He_2^+

Open solution

Q121. JEE Main 2021 (Online) 24th February Evening Shift

The correct set from the following in which both pairs are in correct order of melting point is :

(A) LiCl > LiF ; NaCl > MgO

(B) LiCl > LiF ; MgO > NaCl

(C) LiF > LiCl ; NaCl > MgO

(D) LiF > LiCl ; MgO > NaCl

Open solution

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Q122. JEE Main 2021 (Online) 24th February Evening Shift

The correct shape and $I-I-I$ bond angles respectively in I_3^- ion are :

(A) Linear; 180°

(B) T-shaped; 180° and 90°

(C) Trigonal planar; 120°

(D) Distorted trigonal planar; 135° and 90°

Open solution

Q123. JEE Main 2021 (Online) 24th February Morning Shift

Which of the following are isostructural pairs ?

A. SO_4^{2-} and CrO_4^{2-}

B. $SiCl_4$, and $TiCl_4$

C. NH_3 and NO_3^-

D. BCl_3 and $BrCl_3$

(A) B and C only (B) C and D only (C) A and B only (D) A and C only

Open solution

Q124. JEE Main 2020 (Online) 5th September Evening Slot

The compound that has the largest H–M–H bond angle (M = N, O, S, C) is :

(A) CH_4 (B) H_2S (C) NH_3 (D) H_2O

Open solution

Q125. JEE Main 2020 (Online) 5th September Morning Slot

The structure of PCl_5 in the solid state is :

(A) square pyramidal

(B) tetrahedral $[PCl_4]^+$ and octahedral $[PCl_6]^-$

(C) square planar $[PCl_4]^+$ and octahedral $[PCl_6]^-$

(D) trigonal bipyramidal

Open solution

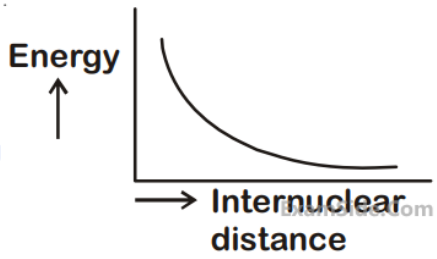
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Q126. JEE Main 2020 (Online) 5th September Morning Slot

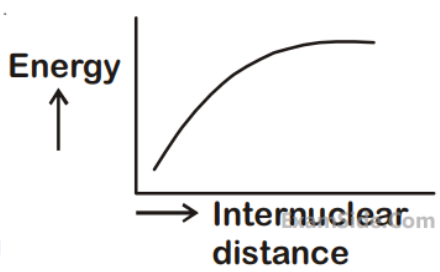
Open solution

The potential energy curve for the H_2 molecule as a function of internuclear distance is :

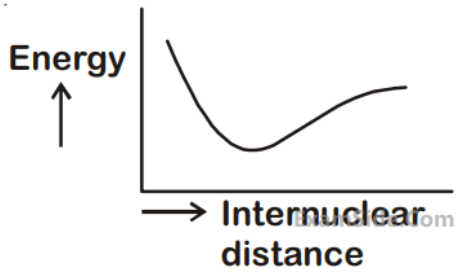
(A)



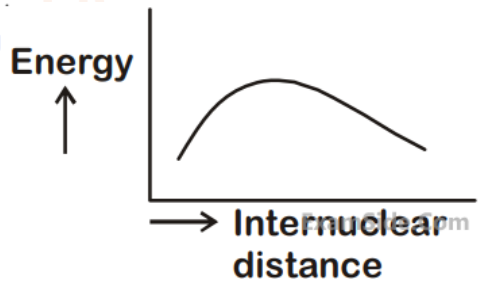
(B)



(C)



(D)



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Q127. JEE Main 2020 (Online) 4th September Evening Slot

Open solution

The reaction in which the hybridisation of the underlined atom is affected is :

(A)

$$\underline{Xe}F_4 + SbF_5 \longrightarrow$$

(B)

$$H_2\underline{S}O_4 + NaCl \xrightarrow{420K}$$

(C)

$$H_3\underline{P}O_2 \xrightarrow{Disproportionation}$$

(D)

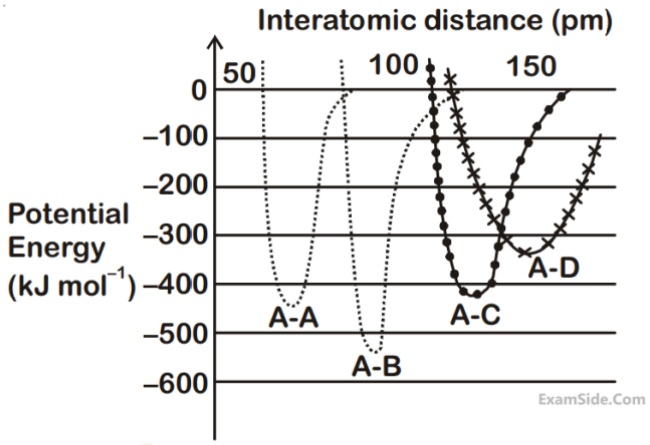
$$\underline{N}H_3 \xrightarrow{H^+}$$

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Q128. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

The intermolecular potential energy for the molecules A, B, C and D given below suggests that :



(A) A-B has the stiffest bond

(B) A-D has the shortest bond length

(C) A-A has the largest bond enthalpy

(D) D is more electronegative than other atoms

Q129. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

Of the species, NO, NO^+, NO^{2+} and NO^-, the one with minimum bond strength is :

(A) NO^- (B) NO (C) NO^+ (D) NO^{2+}

Q130. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

Match the type of interaction in column A with the distance dependence of their interaction energy in column B

A	B
(i) ion-ion	(a) $\frac{1}{r}$
(ii) dipole-dipole	(b) $\frac{1}{r^2}$
(iii) London dispersion	(c) $\frac{1}{r^3}$
	(d) $\frac{1}{r^6}$

(A) (I)-(a), (II)-(b), (III)-(d)

(B) (I)-(b), (II)-(d), (III)-(c)

(C) (I)-(a), (II)-(b), (III)-(c)

(D) (I)-(a), (II)-(c), (III)-(d)

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Q131. JEE Main 2020 (Online) 2nd September Evening Slot

The molecular geometry of SF_6 is octahedral. What is the geometry of SF_4 (including lone pair(s) of electrons, if any)?

(A) Tetrahedral
(B) Trigonal bipyramidal
(C) Square planar
(D) Pyramidal

Open solution

Q132. JEE Main 2020 (Online) 2nd September Evening Slot

The shape / structure of $[XeF_5]^-$ and XeO_3F_2, respectively, are

(A) Pentagonal planar and trigonal bipyramidal
(B) Trigonal bipyramidal and pentagonal planar
(C) Octahedral and square pyramidal
(D) Trigonal bipyramidal and trigonal bipyramidal

Open solution

Q133. JEE Main 2020 (Online) 2nd September Morning Slot

If AB_4 molecule is a polar molecule, a possible geometry of AB_4 is

(A) Tetrahedral
(B) see-saw
(C) Square pyramidal
(D) Square planar

Open solution

Q134. JEE Main 2020 (Online) 9th January Evening Slot

The number of sp^2 hybrid orbitals in a molecule of benzene is :

(A) 24
(B) 12
(C) 6
(D) 18

Open solution

Q135. JEE Main 2020 (Online) 9th January Morning Slot

If the magnetic moment of a dioxygen species is 1.73 B.M, it may be :

(A) O_2^- or O_2^+
(B) O_2, O_2^- or O_2^+
(C) O_2 or O_2^+
(D) O_2 or O_2^-

Open solution

Q136. JEE Main 2020 (Online) 8th January Evening Slot

Arrange the following bonds according to their average bond energies in descending order :
C–Cl, C–Br, C–F, C–I

(A) C–Br > C–I > C–Cl > C–F
(B) C–Cl > C–Br > C–I > C–F
(C) C–I > C–Br > C–Cl > C–F
(D) C–F > C–Cl > C–Br > C–I

Open solution

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Q137. JEE Main 2020 (Online) 7th January Evening Slot

The bond order and the magnetic characteristics of CN^- are :

(A) 3, paramagnetic
(B) $2\frac{1}{2}$, paramagnetic
(C) 3, diamagnetic
(D) $2\frac{1}{2}$, diamagnetic

Open solution

Q138. JEE Main 2020 (Online) 7th January Morning Slot

The relative strength of interionic/intermolecular forces in decreasing order is :

(A) dipole-dipole > ion-dipole > ion-ion
(B) ion-dipole > dipole-dipole > ion-ion
(C) ion-dipole > ion-ion > dipole-dipole
(D) ion-ion > ion-dipole > dipole-dipole

Open solution

Q139. JEE Main 2020 (Online) 7th January Morning Slot

The dipole of CCl_4, $CHCl_3$ and CH_4 are in the order :

(A) $CCl_4 < CH_4 < CHCl_3$
(B) $CH_4 = CCl_4 < CHCl_3$
(C) $CH_4 < CCl_4 < CHCl_3$
(D) $CHCl_3 < CH_4 = CCl_4$

Open solution

Q140. JEE Main 2019 (Online) 12th April Morning Slot

The correct statement among the following is :

(A) $(SiH_3)_3N$ is planar and more basic than $(CH_3)_3N$
(B) $(SiH_3)_3N$ is pyramidal and more basic than $(CH_3)_3N$
(C) $(SiH_3)_3N$ is pyramidal and less basic than $(CH_3)_3N$
(D) $(SiH_3)_3N$ is planar and less basic than $(CH_3)_3N$

Open solution

Q141. JEE Main 2019 (Online) 10th April Morning Slot

During the change of O_2 to O_2^-, the incoming electron goes to the orbital :

(A) $\pi 2p_y$
(B) $\sigma^* 2p_z$
(C) $\pi^* 2p_x$
(D) $\pi 2p_x$

Open solution

Q142. JEE Main 2019 (Online) 9th April Evening Slot

Among the following species, the diamagnetic molecule is

(A) CO
(B) B_2
(C) O_2
(D) NO

Open solution

Q143. JEE Main 2019 (Online) 9th April Evening Slot

HF has highest boiling point among hydrogen halides, because it has :

(A) lowest dissociation enthalpy
(B) strongest hydrogen bonding
(C) lowest ionic character
(D) strongest van der Waals' interactions

Open solution

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Q144. JEE Main 2019 (Online) 9th April Morning Slot

Among the following, the molecule expected to be stabilized by anion formation is :
C_2, O_2, NO, F_2

(A) C_2 (B) NO (C) O_2 (D) F_2

Open solution

Q145. JEE Main 2019 (Online) 8th April Evening Slot

The ion that has sp^3d^2 hybridization for the central atom, is :

(A) $[ICl_2]^-$ (B) $[ICl_4]^-$ (C) $[IF_6]^-$ (D) $[BrF_2]^-$

Open solution

Q146. JEE Main 2019 (Online) 8th April Evening Slot

The covalent alkaline earth metal halide (X = Cl, Br, I) is :

(A) CaX_2 (B) SrX_2 (C) MgX_2 (D) BeX_2

Open solution

Q147. JEE Main 2019 (Online) 8th April Evening Slot

Among the following molecules / ions, C_2^{2-}, N_2^{2-}, O_2^{2-}, O_2 which one is diamagnetic and has the shortest bond length?

(A) O_2 (B) C_2^{2-} (C) N_2^{2-} (D) O_2^{2-}

Open solution

Q148. JEE Main 2019 (Online) 8th April Evening Slot

The correct statement about ICl_5 and ICl_4^- is

(A) ICl_5 is square pyramidal and ICl_4^- is square planar.

(B) ICl_5 is trigonal bipyramidal and ICl_4^- is tetrahedral.

(C) ICl_5 is square pyramidal and ICl_4^- is tetrahedral

(D) Both are isostructural.

Open solution

Q149. JEE Main 2019 (Online) 10th January Morning Slot

Two pi and half sigma bonds are present in :

(A) O_2 (B) N_2^+ (C) O_2^+ (D) N_2

Open solution

Q150. JEE Main 2019 (Online) 10th January Morning Slot

The type of hybridisation and number of lone pair (s) of electrons of Xe in $XeOF_4$, respectively, are:

(A) sp^3d and 2 (B) sp^3d^2 and 2 (C) sp^3d and 1 (D) sp^3d^2 and 1

Open solution

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Q151. JEE Main 2019 (Online) 9th January Evening Slot

In which of the following processes, the bond order has increased and paramagnetic character has changed to diamagnetic ?

(A) $NO \rightarrow NO^+$ (B) $N_2 \rightarrow N_2^+$ (C) $O_2 \rightarrow O_2^+$ (D) $O_2 \rightarrow O_2^{2-}$

Open solution

Q152. JEE Main 2019 (Online) 9th January Morning Slot

According to molecular orbital theory, which of the following is true with respect to Li_2^+ and Li_2^- ?

(A) Li_2^+ is unstable and Li_2^- is stable (B) Li_2^+ is stable and Li_2^- unstable

(C) Both are stable (D) Both are unstable

Open solution

Q153. JEE Main 2018 (Online) 16th April Morning Slot

The **incorrect** geometry is represented by :

(A) BF_3 - trigonal planar (B) H_2O - bent

(C) NF_3 - trigonal planar (D) AsF_5 - trigonal bipyramidal

Open solution

Q154. JEE Main 2018 (Online) 16th April Morning Slot

Which of the following conversions involves change in both shape and hybridisation ?

(A) $NH_3 \rightarrow NH_4^+$ (B) $CH_4 \rightarrow C_2H_6$ (C) $H_2O \rightarrow H_3O^+$ (D) $BF_3 \rightarrow BF_4^-$

Open solution

Q155. JEE Main 2018 (Offline)

According to molecular orbital theory, which of the following will not be a viable molecule?

(A) He_2^{2+} (B) He_2^+ (C) H_2^- (D) H_2^{2-}

Open solution

Q156. JEE Main 2018 (Offline)

Which of the following compounds contain(s) no covalent bond(s)? KCl, PH_3, O_2, B_2H_6, H_2SO_4

(A) KCl, B_2H_6 (B) KCl, B_2H_6, PH_3 (C) KCl, H_2SO_4 (D) KCl

Open solution

Q157. JEE Main 2018 (Offline)

Total number of lone pair of electrons in I_3^- ion is

(A) 3 (B) 9 (C) 6 (D) 12

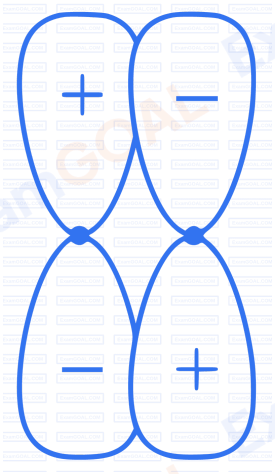
Open solution

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Q158. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

Which of the following best describes the diagram below of a molecular orbital ?



(A) A non-bonding orbital

(B) An antibonding σ orbital

(C) A bonding π orbital

(D) An antibonding π orbital

Q159. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

In the molecular orbital diagram for the molecular ion, N_2^+, the number of electrons in the σ_{2p_z} molecular orbital is :

(A) 0 (B) 1 (C) 2 (D) 3

Q160. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

Identify the pair in which the geometry of the species is T-shape and square - pyramidal, respectively :

(A) ClF_3 and IO_4^- (B) ICl_2^- and ICl_5 (C) $XeOF_2$ and $XeOF_4$ (D) IO_3^- and $IO_2F_2^-$

Q161. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

The decreasing order of bond angles in BF_3, NH_3, PF_3 and I_3^- is :

(A) $I_3^- > NH_3 > PF_3 > BF_3$

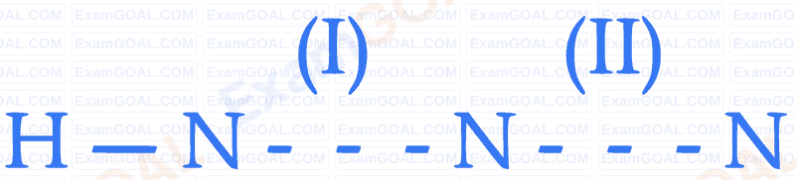
(B) $I_3^- > BF_3 > NH_3 > PF_3$

(C) $BF_3 > I_3^- > PF_3 > NH_3$

(D) $BF_3 > NH_3 > PF_3 > I_3^-$

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Q162. JEE Main 2018 (Online) 15th April Morning Slot



Open solution

In hydrogen azide (above) the bond orders of bond (I) and (II) are :

(A) (I) < 2 (II) > 2

(B) (I) > 2 (II) < 2

(C) (I) > 2 (II) > 2

(D) (I) < 2 (II) < 2

Q163. JEE Main 2017 (Online) 9th April Morning Slot

The group having triangular planar structures is :

(A) BF_3, NF_3, CO_3^{2-}

(B) CO_3^{2-}, NO_3^-, SO_3

(C) NH_3, SO_3, CO_3^{2-}

(D) NCl_3, BCl_3, SO_3

Open solution

Q164. JEE Main 2017 (Online) 8th April Morning Slot

Which of the following is paramagnetic ?

(A) NO^+ (B) CO (C) O_2^{2-} (D) B_2

Open solution

Q165. JEE Main 2017 (Online) 8th April Morning Slot

sp^3d^2 hybridization is **not** displayed by :

(A) BrF_5 (B) SF_6 (C) $[CrF_6]^{3-}$ (D) PF_5

Open solution

Q166. JEE Main 2017 (Offline)

Which of the following species is **not** paramagnetic?

(A) CO (B) O_2 (C) B_2 (D) NO

Open solution

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Q167. JEE Main 2016 (Online) 10th April Morning Slot

The bond angle H - X - H is the greatest in the compound :

(A) CH_4 (B) NH_3 (C) H_2O (D) PH_3

Open solution

Q168. JEE Main 2016 (Online) 9th April Morning Slot

The group of molecules having identical shape is :

(A) SF_4 , XeF_4 , CCl_4 (B) ClF_3 , $XeOF_2$, XeF_3^+

(C) BF_3 , PCl_3 , XeO_3 (D) PCl_5 , IF_5 , XeO_2F_2

Open solution

Q169. JEE Main 2016 (Offline)

The species in which the N atom is in a state of sp hybridization is :

(A) NO_2^+ (B) NO_2^- (C) NO_3^- (D) NO_2

Open solution

Q170. JEE Main 2015 (Online) 11th April Morning Slot

Molecule AB has a bond length of 1.617 $\mathring{A}$ and a dipole moment of 0.38 D. The fractional charge on each atom (absolute magnitude) is : (e_0 = 4.802 $\times$ 10^{-10} esu)

(A) 0 (B) 0.05 (C) 0.5 (D) 1.0

Open solution

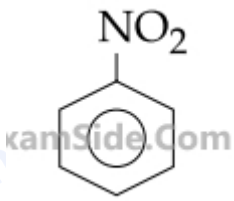
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Q171. JEE Main 2015 (Online) 11th April Morning Slot

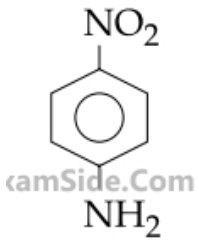
Open solution

Which compound exhibits maximum dipole moment among the following ?

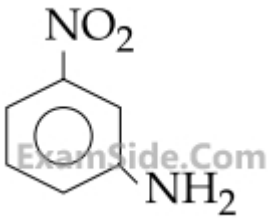
(A)



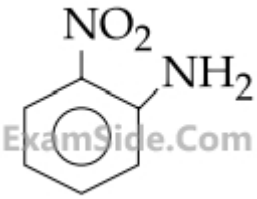
(B)



(C)



(D)



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Q172. JEE Main 2015 (Online) 10th April Morning Slot

After understanding the assertion and reason, choose the correct option.

Open solution

Assertion : In the bonding molecular orbital (MO) of H_2 , electron density is increased between the nuclei.

Reason : The bonding MO is $\psi_A + \psi_B$, which shows destructive interference of the combining electron waves.

(A) Assertion and reason are correct and reason is the correct explanation for the assertion

(B) Assertion and reason are correct, but reason is not the correct explanation for the assertion.

(C) Assertion is correct, reason is incorrect.

(D) Assertion is incorrect, reason is correct.

Q173. JEE Main 2015 (Online) 10th April Morning Slot

The geometry of $XeOF_4$ by VSEPR theory is :

Open solution

(A) trigonal bipyramidal (B) square pyramidal

(C) octahedral (D) pentagonal planar

Q174. JEE Main 2015 (Offline)

The intermolecular interaction that is dependent on the inverse cube of distance between the molecule is:

Open solution

(A) ion-dipole interaction (B) London force

(C) hydrogen bond (D) ion-ion interaction

Q175. JEE Main 2014 (Online) 19th April Morning Slot

Which one of the following molecules is paramagnetic ?

Open solution

(A) N_2 (B) NO (C) CO (D) O_3

Q176. JEE Main 2014 (Online) 19th April Morning Slot

Amongst LiCl, RbCl, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are :

Open solution

(A) LiCl and RbCl (B) RbCl and $BeCl_2$ (C) $MgCl_2$ and $BeCl_2$ (D) RbCl and $MgCl_2$

Q177. JEE Main 2014 (Online) 12th April Morning Slot

Which of the following molecules has two sigma(σ) and two pi(π) bonds ?

Open solution

(A) C_2H_4 (B) N_2F_2 (C) $C_2H_2Cl_2$ (D) HCN

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Q178. JEE Main 2014 (Online) 11th April Morning Slot

The correct order of bond dissociation energy among N_2, O_2, O_2^- is shown in which of the following arrangements ?

(A) $N_2 > O_2^- > O_2$ (B) $O_2^- > O_2 > N_2$ (C) $N_2 > O_2 > O_2^-$ (D) $O_2 > O_2^- > N_2$

Open solution

Q179. JEE Main 2014 (Online) 11th April Morning Slot

Shapes of certain interhalogen compounds are stated below. Which one of them is **not** correctly stated ?

(A) IF_7 : pentagonal bipyramid (B) BrF_5 : trigonal bipyramid

(C) BrF_3 : planar T-shaped (D) ICl_3 : planar dimeric

Open solution

Q180. JEE Main 2014 (Online) 11th April Morning Slot

Which one of the following does **not** have a pyramidal shape ?

(A) $(CH_3)_3N$ (B) $(SiH_3)_3N$ (C) $P(CH_3)_3$ (D) $P(SiH_3)_3$

Open solution

Q181. JEE Main 2014 (Online) 11th April Morning Slot

In allene (C_3H_4), the type(s) of hybridization of the carbon atoms is (are) :

(A) sp and sp^3 (B) sp^2 and sp (C) only sp^2 (D) sp^2 and sp^3

Open solution

Q182. JEE Main 2014 (Online) 9th April Morning Slot

Which of the following has unpaired electron(s) ?

(A) N_2 (B) O_2^- (C) N_2^{2+} (D) O_2^{2-}

Open solution

Q183. JEE Main 2014 (Online) 9th April Morning Slot

The number and type of bonds in C_2^{2-} ion in CaC_2 are :

(A) One σ bond and one π – bond (B) one σ bond and two π – bonds

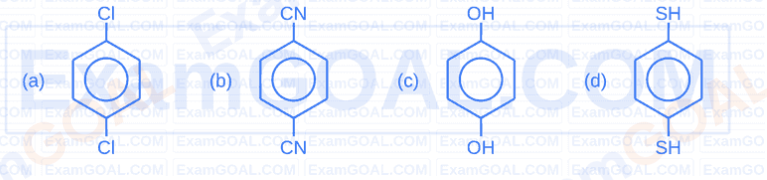
(C) Two σ bonds and two π – bonds (D) Two σ bonds and one π – bond

Open solution

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Q184. JEE Main 2014 (Offline)

For which of the following molecule significant $\mu \neq 0$?



Open solution

(A) Only (a) (B) (a) and (b) (C) Only (c) (D) (c) and (d)

Q185. JEE Main 2013 (Online) 25th April Morning Slot

In which of the following sets, all the given species are isostructural?

(A) CO_2, NO_2, ClO_2, SiO_2 (B) $PCl_3, AlCl_3, BCl_3, SbCl_3$

(C) BF_3, NF_3, PF_3, AlF_3 (D) $BF_4^-, CCl_4, NH_4^+, PCl_4^+$

Open solution

Q186. JEE Main 2013 (Online) 25th April Morning Slot

The internuclear distances in O-O bonds for O_2^+, O_2, O_2^- and O_2^{2-} respectively are :

(A) 1.30 Å, 1.49 Å, 1.12 Å, 1.21 Å

(B) 1.49 Å, 1.21 Å, 1.12 Å, 1.30 Å

(C) 1.21 Å, 1.12 Å, 1.49 Å, 1.30 Å

(D) 1.12 Å, 1.21 Å, 1.30 Å, 1.49 Å

Open solution

Q187. JEE Main 2013 (Online) 23th April Morning Slot

The shape of IF_6^- is :

(A) Trigonally distorted octahedron (B) Pyramidal

(C) Octahedral (D) Square antiprism

Open solution

Q188. JEE Main 2013 (Online) 23th April Morning Slot

Bond distance in HF is 9.17×10^{-11} m. Dipole moment of HF is 6.104×10^{-30} Cm. The percentage ionic character in HF will be : (electron charge = 1.60×10^{-19} C)

(A) 61.0% (B) 38.0% (C) 35.5% (D) 41.5%

Open solution

Q189. JEE Main 2013 (Online) 23th April Morning Slot

The correct order of viscosity of the following liquids will be :

(A) Water < methyl alcohol < dimethyl ether < glycerol

(B) methyl alcohol < glycerol < water < dimethyl ether

(C) dimethyl ether < methyl alcohol < water < glycerol

(D) glycerol < dimethyl ether < water < methyl alcohol

Open solution

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Q190. JEE Main 2013 (Online) 23th April Morning Slot

Which has trigonal bipyramidal shape?

(A) $XeOF_4$ (B) XeO_3 (C) XeO_3F_2 (D) $XeOF_2$

Open solution

Q191. JEE Main 2013 (Online) 22th April Morning Slot

Bond order normally gives idea of stability of a molecular species. All the molecules viz. H_2, Li_2 and B_2 have the same bond order yet they are not equally stable. Their stability order is :

(A) $H_2 > B_2 > Li_2$ (B) $Li_2 > H_2 > B_2$ (C) $Li_2 > B_2 > H_2$ (D) None of these

Open solution

Q192. JEE Main 2013 (Online) 9th April Morning Slot

Which one of the following molecules is polar?

(A) XeF_4 (B) IF_5 (C) SbF_5 (D) CF_4

Open solution

Q193. JEE Main 2013 (Online) 9th April Morning Slot

In which of the following ionization processes the bond energy has increased and also the magnetic behaviour has changed from paramagnetic to diamagnetic?

(A) $NO \rightarrow NO^+$ (B) $N_2 \rightarrow N_2^+$ (C) $C_2 \rightarrow C_2^+$ (D) $O_2 \rightarrow O_2^+$

Open solution

Q194. JEE Main 2013 (Online) 9th April Morning Slot

Trigonal bipyramidal geometry is shown by:

(A) $XeOF_2$ (B) XeO_3F_2 (C) $FXeOSO_2F$ (D) $[XeF_5]^{2-}$

Open solution

Q195. JEE Main 2013 (Offline)

In which of the following pairs of molecules/ions, both the species are not likely to exist?

(A) H_2^+, He_2^{2-} (B) H_2^-, He_2^{2-} (C) H_2^{2+}, He_2 (D) H_2^-, He_2^{2+}

Open solution

Q196. JEE Main 2013 (Offline)

Stability of the species Li_2, Li_2^- and Li_2^+ increases in the order of:

(A) $Li_2 < Li_2^+ < Li_2^-$ (B) $Li_2^+ < Li_2^- < Li_2$ (C) $Li_2^- < Li_2^+ < Li_2$ (D) $Li_2^- < Li_2 < Li_2^+$

Open solution

Q197. AIEEE 2012 (Online) 26th May

Among the following species which two have trigonal pyramidal shape? (I) NI_3 (II) I_3^- (III) SO_3^{2-} (IV) NO_3^-

(A) I and III (B) III and IV (C) I and IV (D) II and III

Open solution

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Q198. AIEEE 2012 (Online) 26th May

Dipole moment is shown by

(A) 1,2-dichlorobenzene (B) trans-2,3-dichloro-2-butene

(C) 1,4-dichlorobenzene (D) trans-1,2-dinitroethene

Open solution

Q199. AIEEE 2012 (Online) 19th May

Among the following, the molecule with the lowest dipole moment is

(A) $CHCl_3$ (B) CH_3Cl (C) CH_2Cl_2 (D) CCl_4

Open solution

Q200. AIEEE 2012 (Online) 19th May

Which of the following has the square planar structure?

(A) XeF_4 (B) NH_4^+ (C) BF_4^- (D) CCl_4

Open solution

Q201. AIEEE 2012 (Online) 12th May

The formation of molecular complex $BF_3 - NH_3$ results in a change in hybridization of boron

(A) from sp^2 to dsp^2 (B) from sp^2 to sp^3 (C) from sp^3 to sp^2 (D) from sp^3 to sp^3d

Open solution

Q202. AIEEE 2012 (Online) 12th May

Although CN^- ion and N_2 molecule are isoelectronic, yet N_2 molecule is chemically inert because of

(A) presence of more number of electrons in bonding orbitals

(B) low bond energy

(C) absence of bond polarity

(D) uneven electron distribution.

Open solution

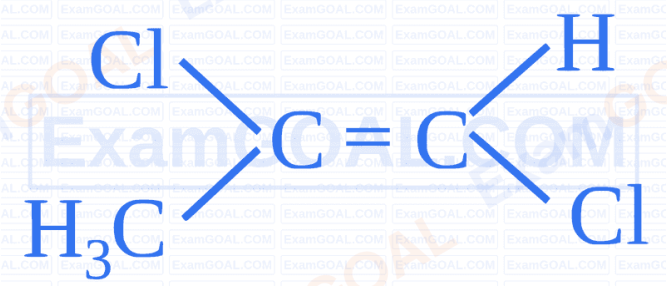
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Q203. AIEEE 2012 (Online) 12th May

Among the following chloro-compound having the lowest dipole moment is

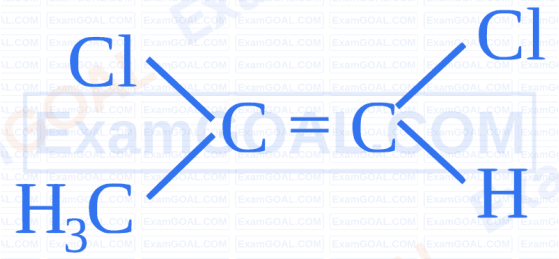
(A) CH_3Cl

(B)



(C) CH_2Cl_2

(D)



Open solution

Q204. AIEEE 2012 (Online) 7th May

Among the following, the species having the smallest bond is

(A) NO^- (B) NO^+ (C) O_2 (D) NO

Open solution

Q205. AIEEE 2012

In which of the following pairs the two species are not isostructural ?

(A) CO_3^{2-} and NO_3^- (B) PCl_4^+ and $SiCl_4$ (C) PF_5 and BrF_5 (D) AlF_6^{3-} and SF_6

Open solution

Q206. AIEEE 2011

Among the following the maximum covalent character is shown by the compound :

(A) $SnCl_2$ (B) $AlCl_3$ (C) $MgCl_2$ (D) $FeCl_2$

Open solution

Q207. AIEEE 2011

The hybridization of orbitals of N atom in NO_3^-, NO_2^+ and NH_4^+ are respectively :

(A) sp , sp^2, sp^3 (B) sp^2, sp , sp^3 (C) sp , sp^3, sp^2 (D) sp^2, sp^3, sp

Open solution

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Q208. AIEEE 2011

The structure of IF_7 is :

(A) trigonal bipyramid
(B) octahedral
(C) pentagonal bipyramid
(D) square pyramid

Open solution

Q209. AIEEE 2009

Using MO theory, predict which of the following species has the shortest bond length?

(A) O_2^+ (B) O_2^- (C) O_2^{2-} (D) O_2^{2+}

Open solution

Q210. AIEEE 2009

The bond dissociation energy of B – F in BF_3 is 646 kJ mol^{-1} whereas that of C – F in CF_4 is 515 kJ mol^{-1}. The correct reason for higher B – F bond dissociation energy as compared to that of C – F is :

(A) smaller size of B-atom as compared to that of C-atom
(B) stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4
(C) significant $p\pi - p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4.
(D) lower degree of $p\pi - p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4.

Open solution

Q211. AIEEE 2008

Which of the following pair of species have the same bond order?

(A) CN^- and NO^+ (B) CN^- and CN^+ (C) O_2^- and CN^- (D) NO^+ and CN^+

Open solution

Q212. AIEEE 2008

The bond dissociation energy of B - F in BF_3 is 646 kJ mol^{-1} whereas that of C - F in CF_4 is 515 kj mol^{-1}. The correct reason for higher B - F bond dissociation energy as compared to that of C - F is

(A) stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4
(B) significant $p\pi - p\pi$ interaction between B and F in BF_3 whereas there is no possibility of such interaction between C and F in CF_4
(C) lower degree of $p\pi - p\pi$ interaction between B and F in BF_3 than that between C and F in CF_4
(D) smaller size of B - atom as compared to that of C- atom.

Open solution

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Q213. AIEEE 2007

Which of the following species exhibits the diamagnetic behaviour?

(A) O_2^{2-} (B) NO (C) O_2^+ (D) O_2

Open solution

Q214. AIEEE 2007

The charge/size ratio of a cation determines its polarizing power. Which one of the following sequences represents the increasing order of the polarizing power of the cationic species, K^+, Ca^{2+}, Mg^{2+}, Be^{2+}?

(A) $Mg^{2+} < Be^{2+} < K^+ < Ca^{2+}$ (B) $K^+ < Ca^{2+} < Mg^{2+} < Be^{2+}$

(C) $Be^{2+} < K^+ < Ca^{2+} < Mg^{2+}$ (D) $Ca^{2+} < Mg^{2+} < Be^{2+} < K^+$

Open solution

Q215. AIEEE 2007

In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed?

(A) $C_2 \rightarrow C_2^+$ (B) $N_2 \rightarrow N_2^+$ (C) $NO \rightarrow NO^+$ (D) $O_2 \rightarrow O_2^+$

Open solution

Q216. AIEEE 2007

Which of the following hydrogen bonds is the strongest?

(A) O–H.....N (B) F–H.....F (C) O–H.....O (D) O–H.....F

Open solution

Q217. AIEEE 2006

Which of the following molecules/ions does not contain unpaired electrons?

(A) O_2^{2-} (B) B_2 (C) N_2^+ (D) O_2

Open solution

Q218. AIEEE 2006

In which of the following molecules/ions are all the bonds **not** equal?

(A) XeF_4 (B) BF_4^- (C) SF_4 (D) SiF_4

Open solution

Q219. AIEEE 2006

The decreasing values of bond angles from NH_3 (106°) to SbH_3 (91°) down group-15 of the periodic table is due to

(A) increasing bp-bp repulsion (B) increasing p-orbital character in sp^3

(C) decreasing lp-bp repulsion (D) decreasing electronegativity

Open solution

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Q220. AIEEE 2006

Among the following mixtures, dipole-dipole as the major interaction, is present in

(A) benzene and ethanol
(B) acetonitrile and acetone
(C) KCl and water
(D) benzene and carbon tetrachloride

Open solution

Q221. AIEEE 2005

Lattice energy of an ionic compound depends upon

(A) Charge on the ion only
(B) Size of the ion only
(C) Packing of ions only
(D) Charge on the ion and size of the ion

Open solution

Q222. AIEEE 2005

Which one of the following species is diamagnetic in nature?

(A) He_2^+
(B) H_2
(C) H_2^+
(D) H_2^-

Open solution

Q223. AIEEE 2005

The molecular shapes of SF_4, CF_4 and XeF_4 are

(A) the same with 2,0 and 1 lone pairs of electrons on the central atom, respectively
(B) the same with 1, 1 and 1 lone pair of electrons on the central atoms, respectively
(C) different with 0, 1 and 2 lone pair of electrons on the central atoms, respectively
(D) different with 1, 0 and 2 lone pairs of electron on the central atoms respectively

Open solution

Q224. AIEEE 2005

The number and type of bonds between two carbon atoms in calcium carbide are

(A) One sigma, one pi
(B) One sigma, two pi
(C) Two sigma, one pi
(D) Two sigma, two pi

Open solution

Q225. AIEEE 2004

The correct order of bond angles (smallest first) in H_2S, NH_3, BF_3 and SiH_4 is

(A) $H_2S < SiH_4 < NH_3 < BF_3$
(B) $H_2S < NH_3 < BF_3 < SiH_4$
(C) $H_2S < NH_3 < SiH_4 < BF_3$
(D) $NH_3 < H_2S < SiH_4 < BF_3$

Open solution

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Q226. AIEEE 2004

The bond order in NO is 2.5 while that in NO^+ is 3. Which of the following statements is true for these two species?

(A) Bond length in NO^+ is greater than in NO

(B) Bond length is unpredictable

(C) Bond length in NO^+ is equal to that in NO

(D) Bond length in NO is greater than in NO^+

Open solution

Q227. AIEEE 2004

The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively

(A) sp^2 and sp^2 (B) sp^3 and sp^3 (C) sp^3 and sp^2 (D) sp^2 and sp^3

Open solution

Q228. AIEEE 2004

Which one of the following has the regular tetrahedral structure?
(Atomic nos : B = 5, S = 16, Ni = 28, Xe = 54)

(A) XeF_4 (B) $[Ni(CN)_4]^{2-}$ (C) BF_4^- (D) SF_4

Open solution

Q229. AIEEE 2004

The maximum number of 90° angles between bond pair of electrons is observed in

(A) dsp^3 hybridization (B) sp^3d^2 hybridization

(C) dsp^2 hybridization (D) sp^3d hybridization

Open solution

Q230. AIEEE 2004

Which one the following does not have sp^2 hybridized carbon?

(A) Acetone (B) Acetamide (C) Acetonitrile (D) Acetic acid

Open solution

Q231. AIEEE 2003

Which one of the following pairs of molecules will have permanent dipole moments for both members

(A) NO_2 and CO_2 (B) NO_2 and O_3 (C) SiF_4 and CO_2 (D) SiF_4 and NO_2

Open solution

Q232. AIEEE 2003

Which one of the following compounds has the smallest bond angle in its molecule?

(A) OH_2 (B) SH_2 (C) NH_3 (D) SO_2

Open solution

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Q233. AIEEE 2003

The pair of species having identical shapes for molecules of both species is

(A) XeF_2, CO_2 (B) BF_3, PCl_3 (C) PF_5, IF_5 (D) CF_4, SF_4

Open solution

Q234. AIEEE 2002

In which of the following species the interatomic bond angle is $109^o28'$?

(A) NH_3, BF_4^- (B) NH_4^+, BF_4^- (C) NH_3, BF_3 (D) NH_2^{-1}, BF_3

Open solution

Q235. AIEEE 2002

Which of the following are arranged in an increasing order of their bond strengths?

(A) $O_2^- < O_2 < O_2^+ < O_2^{2-}$ (B) $O_2^{2-} < O_2^- < O_2 < O_2^+$

(C) $O_2^- < O_2^{2-} < O_2 < O_2^+$ (D) $O_2^+ < O_2 < O_2^- < O_2^{2-}$

Open solution

Q236. AIEEE 2002

Hybridisation of underlined atom changes in

(A) $\underline{Al}H_3$ changes to AlH_4^- (B) $H_2\underline{O}$ changes to H_3O^+

(C) $\underline{N}H_3$ changes to NH_4^+ (D) in all cases

Open solution

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Q237. AIEEE 2002

In which of the following species is the underlined carbon having sp^3 hybridisation?

Open solution

(A) $\underline{C}H_3COOH$

(B) $\underline{C}H_3CH_2OH$

(C) $CH_3\underline{C}OCH_3$

(D) $CH_2=\underline{C}H-CH_3$

Q238. AIEEE 2002

Which of the following statements is true?

Open solution

(A) HF is less polar than HBr

(B) absolutely pure water does not contain any ions

(C) chemical bond formation takes place when forces of attraction overcome the forces of repulsion

(D) in covalency transference of electron takes place

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Numerical 64 questions

Q239. JEE Main 2026 (Online) 5th April Morning Shift

Consider the following species:

$BrF_5, XeF_5^-, BF_4^-, ICl_4^-, XeF_4, SF_4, NH_4^+, ClF_3, XeF_2, ICl_2^-$

Number of species having sp^3d hybridized central atom is ____ .

Open solution

Q240. JEE Main 2026 (Online) 4th April Morning Shift

According to Lewis theory, the total number of σ bond-pairs and lone pair of electrons around the central atom of XeO_6^{4-} ion is ____ .

Open solution

Q241. JEE Main 2025 (Online) 29th January Evening Shift

Total number of non-bonded electrons present in NO_2^- ion based on Lewis theory is _____.

Open solution

Q242. JEE Main 2025 (Online) 22nd January Morning Shift

The number of molecules/ions that show linear geometry among the following is _____.

$SO_2, BeCl_2, CO_2, N_3^-, NO_2, F_2O, XeF_2, NO_2^+, I_3^-, O_3$

Open solution

Q243. JEE Main 2024 (Online) 9th April Evening Shift

Total number of electrons present in (π^*) molecular orbitals of O_2, O_2^+ and O_2^- is _____.

Open solution

Q244. JEE Main 2024 (Online) 9th April Morning Shift

The total number of species from the following in which one unpaired electron is present, is _____.

$N_2, O_2, C_2^-, O_2^-, O_2^{2-}, H_2^+, CN^-, He_2^+$

Open solution

Q245. JEE Main 2024 (Online) 8th April Evening Shift

Number of molecules having bond order 2 from the following molecules is _____.

$C_2, O_2, Be_2, Li_2, Ne_2, N_2, He_2$

Open solution

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Q246. JEE Main 2024 (Online) 8th April Morning Shift

Number of molecules from the following which are exceptions to octet rule is _____.

$CO_2, NO_2, H_2SO_4, BF_3, CH_4, SiF_4, ClO_2, PCl_5, BeF_2, C_2H_6, CHCl_3, CBr_4$

Open solution

Q247. JEE Main 2024 (Online) 6th April Evening Shift

Consider the following reactions

$$NiS + HNO_3 + HCl \rightarrow A + NO + S + H_2O$$

$$A + NH_4OH + H_3C - \underset{\displaystyle H_3C - C = N - OH}{\overset{\displaystyle}{\underset{|}{C}}} = N - OH \longrightarrow B + NH_4Cl + H_2O$$

The number of protons that do not involve in hydrogen bonding in the product B is _____.

Open solution

Q248. JEE Main 2024 (Online) 6th April Evening Shift

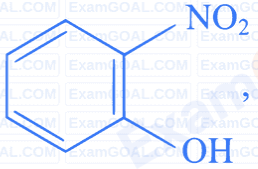
Total number of species from the following with central atom utilising sp^2 hybrid orbitals for bonding is _____.

$NH_3, SO_2, SiO_2, BeCl_2, C_2H_2, C_2H_4, BCl_3, HCHO, C_6H_6, BF_3, C_2H_4Cl_2$

Open solution

Q249. JEE Main 2024 (Online) 6th April Morning Shift

Number of molecules from the following which can exhibit hydrogen bonding is _____. (nearest integer)

$CH_3OH, H_2O, C_2H_6, C_6H_6,$  $, HF, NH_3$

Open solution

Q250. JEE Main 2024 (Online) 5th April Evening Shift

Number of compounds from the following with zero dipole moment is _____.

$HF, H_2, H_2S, CO_2, NH_3, BF_3, CH_4, CHCl_3, SiF_4, H_2O, BeF_2$

Open solution

Q251. JEE Main 2024 (Online) 5th April Morning Shift

In the lewis dot structure for NO_2^-, total number of valence electrons around nitrogen is _____.

Open solution

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Q252. JEE Main 2024 (Online) 4th April Evening Shift

Number of compounds / species from the following with non-zero dipole moment is _____.

$BeCl_2, BCl_3, NF_3, XeF_4, CCl_4, H_2O, H_2S, HBr, CO_2, H_2, HCl$

Open solution

Q253. JEE Main 2024 (Online) 4th April Morning Shift

Number of molecules/species from the following having one unpaired electron is _____.

$O_2, O_2^{-1}, NO, CN^{-1}, O_2^{2-}$

Open solution

Q254. JEE Main 2024 (Online) 1st February Morning Shift

The number of molecules/ion/s having trigonal bipyramidal shape is _____.

$PF_5, BrF_5, PCl_5, [PtCl_4]^{2-}, BF_3, Fe(CO)_5$

Open solution

Q255. JEE Main 2024 (Online) 1st February Morning Shift

The lowest oxidation number of an atom in a compound A_2B is -2 . The number of electrons in its valence shell is _____.

Open solution

Q256. JEE Main 2024 (Online) 31st January Evening Shift

A diatomic molecule has a dipole moment of $1.2\,D$. If the bond distance is $1\,A^\circ$, then fractional charge on each atom is _____ $\times 10^{-1}$ esu.

(Given $1\,D = 10^{-18}$ esucm)

Open solution

Q257. JEE Main 2024 (Online) 31st January Morning Shift

The number of species from the following in which the central atom uses sp^3 hybrid orbitals in its bonding is _____.

$NH_3, SO_2, SiO_2, BeCl_2, CO_2, H_2O, CH_4, BF_3$

Open solution

Q258. JEE Main 2024 (Online) 30th January Morning Shift

The total number of molecular orbitals formed from $2s$ and $2p$ atomic orbitals of a diatomic molecule is _____.

Open solution

Q259. JEE Main 2024 (Online) 29th January Evening Shift

The total number of molecules with zero dipole moment among $CH_4, BF_3, H_2O, HF, NH_3, CO_2$ and SO_2 is _____.

Open solution

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Q260. JEE Main 2024 (Online) 29th January Evening Shift

The total number of anti bonding molecular orbitals, formed from $2s$ and $2p$ atomic orbitals in a diatomic molecule is _____.

Open solution

Q261. JEE Main 2024 (Online) 29th January Morning Shift

The number of species from the following which are paramagnetic and with bond order equal to one is _____.

$H_2, He_2^+, O_2^+, N_2^{2-}, O_2^{2-}, F_2, Ne_2^+, B_2$

Open solution

Q262. JEE Main 2024 (Online) 29th January Morning Shift

Number of compounds with one lone pair of electrons on central atom amongst following is _____.

$O_3, H_2O, SF_4, ClF_3, NH_3, BrF_5, XeF_4$

Open solution

Q263. JEE Main 2024 (Online) 27th January Evening Shift

The number of non-polar molecules from the following is _____.

$HF, H_2O, SO_2, H_2, CO_2, CH_4, NH_3, HCl, CHCl_3, BF_3$

Open solution

Q264. JEE Main 2024 (Online) 27th January Morning Shift

Sum of bond order of CO and NO^+ is _____.

Open solution

Q265. JEE Main 2023 (Online) 11th April Evening Shift

The maximum number of lone pairs of electrons on the central atom from the following species is _____.

ClO_3^-, XeF_4, SF_4 and I_3^-

Open solution

Q266. JEE Main 2023 (Online) 10th April Evening Shift

The number of molecules from the following which contain only two lone pair of electrons is _____

$H_2O, N_2, CO, XeF_4, NH_3, NO, CO_2, F_2$

Open solution

Q267. JEE Main 2023 (Online) 10th April Morning Shift

The number of bent-shaped molecule/s from the following is _____

$N_3^-, NO_2^-, I_3^-, O_3, SO_2$

Open solution

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Q268. JEE Main 2023 (Online) 10th April Morning Shift

The sum of lone pairs present on the central atom of the interhalogen IF_5 and IF_7 is _____

Open solution

Q269. JEE Main 2023 (Online) 8th April Evening Shift

The number of species from the following carrying a single lone pair on central atom Xenon is _____.

$XeF_5^+, XeO_3, XeO_2F_2, XeF_5^-, XeO_3F_2, XeOF_4, XeF_4$

Open solution

Q270. JEE Main 2023 (Online) 8th April Morning Shift

The number of following factors which affect the percent covalent character of the ionic bond is _____

(A) Polarising power of cation

(B) Extent of distortion of anion

(C) Polarisability of the anion

(D) Polarising power of anion

Open solution

Q271. JEE Main 2023 (Online) 6th April Evening Shift

The number of species having a square planar shape from the following is _____.

$XeF_4, SF_4, SiF_4, BF_4^-, BrF_4^-, [Cu(NH_3)_4]^{2+}, [FeCl_4]^{2-}, [PtCl_4]^{2-}$

Open solution

Q272. JEE Main 2023 (Online) 6th April Evening Shift

In an ice crystal, each water molecule is hydrogen bonded to _____ neighbouring molecules.

Open solution

Q273. JEE Main 2023 (Online) 6th April Morning Shift

The number of species from the following which have square pyramidal structure is _____

$PF_5, BrF_4^-, IF_5, BrF_5, XeOF_4, ICl_4^-$

Open solution

Q274. JEE Main 2023 (Online) 31st January Evening Shift

Amongst the following, the number of species having the linear shape is _____.

$XeF_2, I_3^+, C_3O_2, I_3^-, CO_2, SO_2, BeCl_2$ and $BCl_2^{\ominus}$

Open solution

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Q275. JEE Main 2023 (Online) 29th January Morning Shift

The number of molecules or ions from the following, which do not have odd number of electrons are _____.

(A) NO_2

(B) ICl_4^-

(C) BrF_3

(D) ClO_2

(E) NO_2^+

(F) NO

Open solution

Q276. JEE Main 2023 (Online) 25th January Morning Shift

The total number of lone pairs of electrons on oxygen atoms of ozone is _____.

Open solution

Q277. JEE Main 2022 (Online) 30th July Evening Shift

On the basis of MO. theory, the number of molecules from the following which have bond order two, is _____.

$O_2, O_2^+, N_2^{2-}, C_2, B_2$

Open solution

Q278. JEE Main 2022 (Online) 30th July Evening Shift

Consider the molecules / ions given below

$XeO_3, BF_4^-, I_3^-, SF_6, PCl_5$

The ratio of number of molecules having sp^3d hybridisation to those having sp^3 hybridisation is _____

Open solution

Q279. JEE Main 2022 (Online) 29th July Evening Shift

Consider, $PF_5, BrF_5, PCl_3, SF_6, [ICl_4]^-, ClF_3$ and IF_5.

Amongst the above molecule(s)/ion(s), the number of molecule(s)/ion(s) having $sp^3 d^2$ hybridisation is _____.

Open solution

Q280. JEE Main 2022 (Online) 28th July Morning Shift

The number of paramagnetic species among the following is _____.

$B_2, Li_2, C_2, C_2^-, O_2^{2-}, O_2^+$ and He_2^+

Open solution

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Q281. JEE Main 2022 (Online) 28th July Morning Shift

The number of interhalogens from the following having square pyramidal structure is :

$ClF_3, IF_7, BrF_5, BrF_3, I_2Cl_6, IF_5, ClF, ClF_5$

Open solution

Q282. JEE Main 2022 (Online) 27th July Evening Shift

The number of molecule(s) or ion(s) from the following having non-planar structure is _____.

$NO_3^-, H_2O_2, BF_3, PCl_3, XeF_4, SF_4, XeO_3, PH_4^+, SO_3, [Al(OH)_4]^-$

Open solution

Q283. JEE Main 2022 (Online) 27th July Morning Shift

Amongst the following, the number of oxide(s) which are paramagnetic in nature is

$Na_2O, KO_2, NO_2, N_2O, ClO_2, NO, SO_2, Cl_2O$

Open solution

Q284. JEE Main 2022 (Online) 27th July Morning Shift

According to MO theory, number of species/ions from the following having identical bond order is _____.

$CN^-, NO^+, O_2, O_2^+, O_2^{2+}$

Open solution

Q285. JEE Main 2022 (Online) 25th July Evening Shift

The sum of number of lone pairs of electrons present on the central atoms of XeO_3, $XeOF_4$ and XeF_6, is _____

Open solution

Q286. JEE Main 2022 (Online) 25th July Morning Shift

Among the following species

$N_2, N_2^+, N_2^-, N_2^{2-}, O_2, O_2^+, O_2^-, O_2^{2-}$

the number of species showing diamagnetism is _____.

Open solution

Q287. JEE Main 2022 (Online) 30th June Morning Shift

Amongst the following, the number of molecule/(s) having net resultant dipole moment is _____.

NF_3, BF_3, BeF_2, $CHCl_3$, H_2S, SiF_4, CCl_4, PF_5

Open solution

Q288. JEE Main 2022 (Online) 28th June Morning Shift

The hybridization of P exhibited in PF_5 is sp^xd^y. The value of y is _____.

Open solution

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Q289. JEE Main 2022 (Online) 26th June Evening Shift

Amongst SF_4, XeF_4, CF_4 and H_2O, the number of species with two lone pairs of electrons is _____.

Open solution

Q290. JEE Main 2022 (Online) 25th June Evening Shift

Amongst BeF_2, BF_3, H_2O, NH_3, CCl_4 and HCl, the number of molecules with non-zero net dipole moment is _____.

Open solution

Q291. JEE Main 2021 (Online) 1st September Evening Shift

The spin-only magnetic moment value of B_2^+ species is _____ $\times 10^{-2}$ BM. (Nearest integer)
[Given : $\sqrt{3}$ = 1.73]

Open solution

Q292. JEE Main 2021 (Online) 31st August Evening Shift

According to molecular orbital theory, the number of unpaired electron(s) in O_2^{2-} is :

Open solution

Q293. JEE Main 2021 (Online) 31st August Morning Shift

The number of hydrogen bonded water molecule(s) associated with stoichiometry $CuSO_4.5H_2O$ is _____.

Open solution

Q294. JEE Main 2021 (Online) 27th August Evening Shift

The number of species having non-pyramidal shape among the following is _____.

(A) SO_3

(B) NO_3^-

(C) PCl_3

(D) CO_3^{2-}

Open solution

Q295. JEE Main 2021 (Online) 26th August Morning Shift

AB_3 is an interhalogen T-shaped molecule. The number of lone pairs of electrons on A is _____. (Integer answer)

Open solution

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Q296. JEE Main 2021 (Online) 27th July Evening Shift

The total number of electrons in all bonding molecular orbitals of O_2^{2-} is _____.

(Round off to the nearest integer)

Open solution

Q297. JEE Main 2021 (Online) 27th July Morning Shift

The difference between bond orders of CO and $NO^{\oplus}$ is $\frac{x}{2}$ where x = _____. (Round off to the Nearest Integer)

Open solution

Q298. JEE Main 2021 (Online) 20th July Morning Shift

The number of lone pairs of electrons on the central I atom in I_3^- is _____.

Open solution

Q299. JEE Main 2021 (Online) 18th March Evening Shift

The number of species below that have two lone pairs of electrons in their central atom is _____. (Round off to the Nearest Integer).

SF_4, BF_4^-, ClF_3, AsF_3, PCl_5, BrF_5, XeF_4, SF_6

Open solution

Q300. JEE Main 2021 (Online) 18th March Morning Shift

AX is a covalent diatomic molecule where A and X are second row elements of periodic table. Based on Molecular orbital theory, the bond order of AX is 2.5. The total number of electrons in AX is _____. (Round off to the Nearest Integer).

Open solution

Q301. JEE Main 2021 (Online) 25th February Morning Shift

Among the following, the number of halide(s) which is/are inert to hydrolysis is _____.

(A) BF_3

(B) $SiCl_4$

(C) PCl_5

(D) SF_6

Open solution

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Q302. JEE Main 2020 (Online) 7th January Morning Slot

Chlorine reacts with hot and concentrated NaOH and produces compounds (X) and (Y). Compound (X) gives white precipitate with silver nitrate solution. The average bond order between Cl and O atoms in (Y) is.

Open solution

MCQ (More than One Correct Answer) 1 questions

Q303. JEE Main 2013 (Offline)

Which one of the following molecules is expected to exhibit diamagnetic behaviour?

(A) N_2 (B) O_2 (C) S_2 (D) C_2

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **A**	2. **D**	3. **C**	4. **B**	5. **A**
6. **C**	7. **B**	8. **C**	9. **C**	10. **A**
11. **D**	12. **A**	13. **C**	14. **C**	15. **B**
16. **D**	17. **A**	18. **C**	19. **D**	20. **D**
21. **C**	22. **D**	23. **D**	24. **D**	25. **C**
26. **B**	27. **C**	28. **D**	29. **D**	30. **B**
31. **D**	32. **A**	33. **B**	34. **C**	35. **C**
36. **A**	37. **C**	38. **C**	39. **C**	40. **C**
41. **A**	42. **D**	43. **B**	44. **B**	45. **B**
46. **B**	47. **B**	48. **D**	49. **B**	50. **C**
51. **D**	52. **A**	53. **C**	54. **D**	55. **A**
56. **C**	57. **A**	58. **D**	59. **D**	60. **C**
61. **D**	62. **B**	63. **B**	64. **C**	65. **B**
66. **D**	67. **C**	68. **C**	69. **C**	70. **A**
71. **D**	72. **B**	73. **D**	74. **B**	75. **B**
76. **B**	77. **B**	78. **B**	79. **C**	80. **A**
81. **D**	82. **A**	83. **B**	84. **C**	85. **B**
86. **D**	87. **B**	88. **B**	89. **B**	90. **C**
91. **A**	92. **B**	93. **C**	94. **A**	95. **B**
96. **C**	97. **A**	98. **C**	99. **C**	100. **B**
101. **A**	102. **C**	103. **C**	104. **B**	105. **A**
106. **D**	107. **C**	108. **A**	109. **C**	110. **D**
111. **C**	112. **D**	113. **C**	114. **B**	115. **A**
116. **C**	117. **A**	118. **C**	119. **D**	120. **B**
121. **D**	122. **A**	123. **C**	124. **A**	125. **B**
126. **C**	127. **A**	128. **A**	129. **A**	130. **D**
131. **B**	132. **A**	133. **B**	134. **D**	135. **A**
136. **D**	137. **C**	138. **D**	139. **B**	140. **D**
141. **C**	142. **A**	143. **B**	144. **A**	145. **B**
146. **D**	147. **B**	148. **A**	149. **B**	150. **D**

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151. **A**	152. **C**	153. **C**	154. **D**	155. **D**
156. **D**	157. **B**	158. **D**	159. **B**	160. **C**
161. **B**	162. **A**	163. **B**	164. **D**	165. **D**
166. **A**	167. **A**	168. **B**	169. **A**	170. **B**
171. **B**	172. **C**	173. **B**	174. **C**	175. **B**
176. **B**	177. **D**	178. **C**	179. **B**	180. **B**
181. **B**	182. **B**	183. **B**	184. **D**	185. **D**
186. **D**	187. **A**	188. **D**	189. **C**	190. **C**
191. **A**	192. **B**	193. **A**	194. **B**	195. **C**
196. **C**	197. **A**	198. **A**	199. **D**	200. **A**
201. **B**	202. **C**	203. **B**	204. **B**	205. **C**
206. **B**	207. **B**	208. **C**	209. **D**	210. **C**
211. **A**	212. **B**	213. **A**	214. **B**	215. **C**
216. **B**	217. **A**	218. **C**	219. **C**	220. **B**
221. **D**	222. **B**	223. **D**	224. **B**	225. **C**
226. **D**	227. **D**	228. **C**	229. **B**	230. **C**
231. **B**	232. **B**	233. **A**	234. **B**	235. **B**
236. **A**	237. **B**	238. **C**		

Numerical

239. **4**	240. **6**	241. **12**	242. **6**	243. **6**
244. **4**	245. **2**	246. **6**	247. **12**	248. **6**
249. **5**	250. **6**	251. **8**	252. **5**	253. **2**
254. **3**	255. **6**	256. **0**	257. **3**	258. **8**
259. **3**	260. **4**	261. **1**	262. **4**	263. **4**
264. **6**	265. **3**	266. **3**	267. **3**	268. **1**
269. **4**	270. **3**	271. **4**	272. **4**	273. **3**
274. **5**	275. **3**	276. **6**	277. **3**	278. **1**
279. **4**	280. **4**	281. **3**	282. **6**	283. **4**
284. **3**	285. **3**	286. **2**	287. **3**	288. **1**
289. **1**	290. **3**	291. **173**	292. **0**	293. **1**
294. **3**	295. **2**	296. **10**	297. **0**	298. **3**
299. **2**	300. **15**	301. **1**	302. **1.66 to 1.67**	

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MCQ (More than One Correct Answer)

303. **A, D**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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